MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BIT ELECTRONICS MODULE, KEY CODE CWH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Built-In-Test (BIT) electronics module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.985 failures per million operating hours for class I or 5.093 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CWH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	-----Module shall pass all remaining tests-----						
D1-D16	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D17	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D18,D19	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D20	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D21-D33	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D34	Ibd	V	S	-0.02	0.22	-0.02	0.22	-0.02	0.22	mA
D35-D50	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
DS1,DS2	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
DS3	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	mA
DS4	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
DS5-DS7	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
DS8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
DS9	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D60-D62	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D63	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D64	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D65	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
E1-E12	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E13,E14	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E15	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E16-E23	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
F1-F16	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F17	Iih	VII	X	-1.2	0.08	-1.2	0.08	-1.2	0.08	mA
F18,F19	Iih	VII	X	-0.8	0.01	-0.8	0.01	-0.8	0.01	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F20	Ith	VII	X	-1.75	-0.59	-1.75	-0.59	-1.75	-0.59	mA
F21-F32	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F33	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F34	Ith	VII	X	-2.35	-0.57	-2.35	-0.57	-2.35	-0.57	mA
F35-F40	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F41	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F42-F50	Ith	VII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
F51,F52	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F53	Ith	VII	X	-3.2	0.08	-3.2	0.08	-3.2	0.08	mA
F54	Ith	VII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
F55	Ith	VII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
F56,F57	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F58	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F59	Ith	VII	X	-1.75	-0.59	-1.75	-0.59	-1.75	-0.59	mA
F60-F62	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F63	Ith	VII	X	-2.4	0.04	-2.4	0.04	-2.4	0.04	mA
F64	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F65	Ith	VII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
G1-G16	Ith	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G17	Ith	VIII	X	-1.2	0.08	-1.2	0.08	-1.2	0.08	mA
G18,G19	Ith	VIII	X	-0.8	0.01	-0.8	0.01	-0.8	0.01	mA
G20	Ith	VIII	X	-1.75	-0.59	-1.75	-0.59	-1.75	-0.59	mA
G21-G32	Ith	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G33	Ith	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G34	Ith	VIII	X	-2.35	-0.57	-2.35	-0.57	-2.35	-0.57	mA
G35-G40	Ith	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G41	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G42-G50	Ith	VIII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
G51,G52	Ith	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G53	Ith	VIII	X	-3.2	0.08	-3.2	0.08	-3.2	0.08	mA
G54	Ith	VIII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
G55	Ith	VIII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
G56,G57	Ith	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G58	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G59	Iil	VIII	X	-1.75	-0.59	-1.75	-0.59	-1.75	-0.59	mA
G60-G62	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G63	Iil	VIII	X	-2.4	0.04	-2.4	0.04	-2.4	0.04	mA
G64	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G65	Iil	VIII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
H1-H23	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J24	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
L1-L36	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L37	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L38-L46	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L47-L50	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L51	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L52,L53	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L54	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L55-L60	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	7	13	7	13	7	13	uF
M2	PSFC	XIII	S	6	12	6	12	6	12	uF
N	Icc	XIV	X	465	1116	465	1116	465	1116	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1	Cin	XVIII	Q	2	50	2	50	2	50	pF
R2-R4	Cin	XVIII	Q	2	75	2	75	2	75	pF
R5	Cin	XVIII	Q	2	50	2	50	2	50	pF
R6-R16	Cin	XVIII	Q	2	75	2	75	2	75	pF
R17	Cin	XVIII	Q	8	100	8	100	8	100	pF
R18,R19	Cin	XVIII	Q	2	75	2	75	2	75	pF
R20	Cin	XVIII	Q	2	150	2	150	2	150	pF
R21-R33	Cin	XVIII	Q	2	75	2	75	2	75	pF
R34	Cin	XVIII	Q	4	100	4	100	4	100	pF
R35-R43	Cin	XVIII	Q	2	75	2	75	2	75	pF
R44	Cin	XVIII	Q	2	50	2	50	2	50	pF
R45-R52	Cin	XVIII	Q	2	75	2	75	2	75	pF
R53	Cin	XVIII	Q	4	150	4	150	4	150	pF
R54	Cin	XVIII	Q	6	100	6	100	6	100	pF
R55	Cin	XVIII	Q	4	100	4	100	4	100	pF
R56,R57	Cin	XVIII	Q	4	75	4	75	4	75	pF
R58	Cin	XVIII	Q	2	75	2	75	2	75	pF
R59	Cin	XVIII	Q	4	75	4	75	4	75	pF
R60-R62	Cin	XVIII	Q	2	75	2	75	2	75	pF
R63	Cin	XVIII	Q	4	100	4	100	4	100	pF
R64	Cin	XVIII	Q	2	75	2	75	2	75	pF
R65	Cin	XVIII	Q	6	100	6	100	6	100	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/520-1 for class I and M28787/520-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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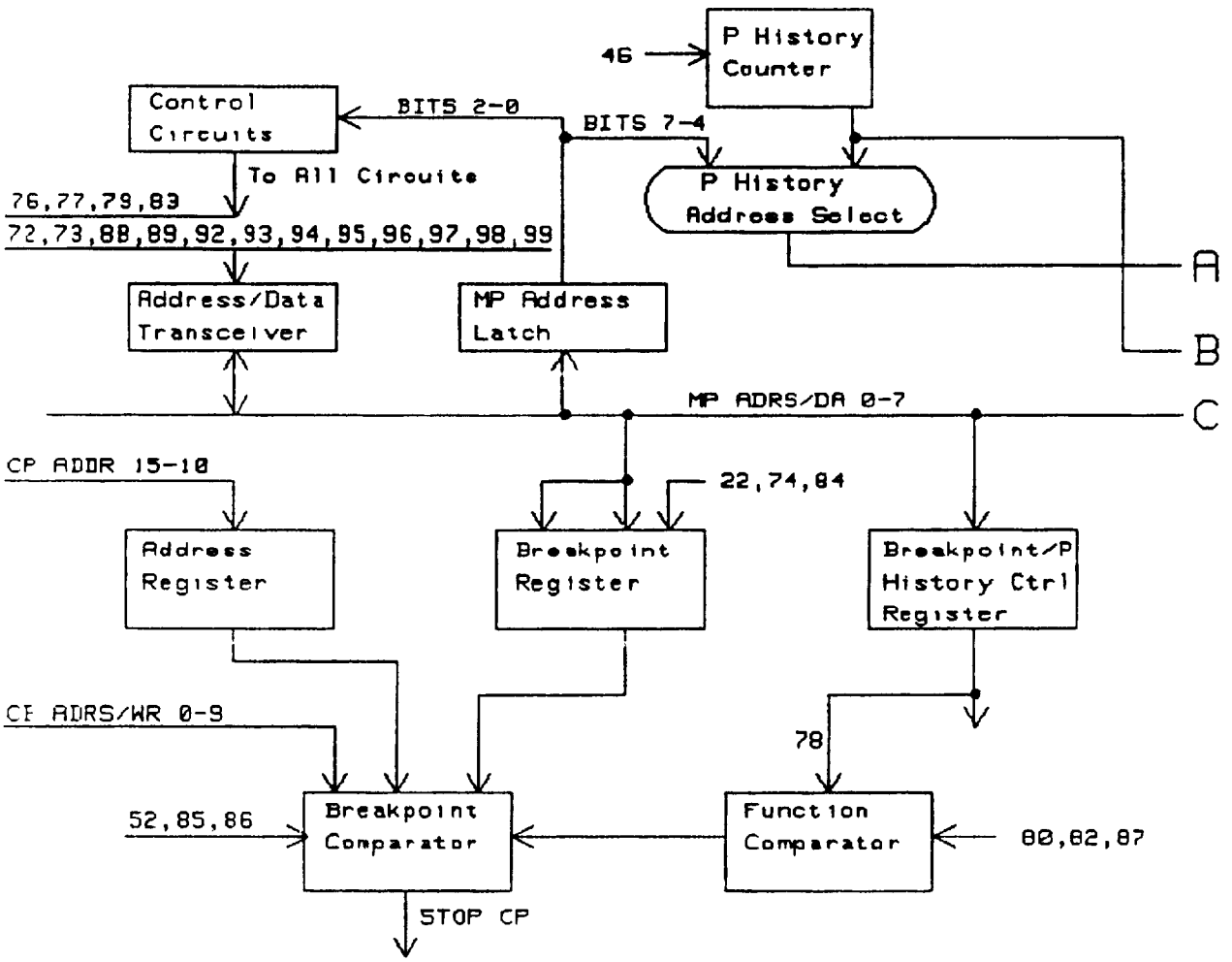


FIGURE 1. CWH functional block diagram.

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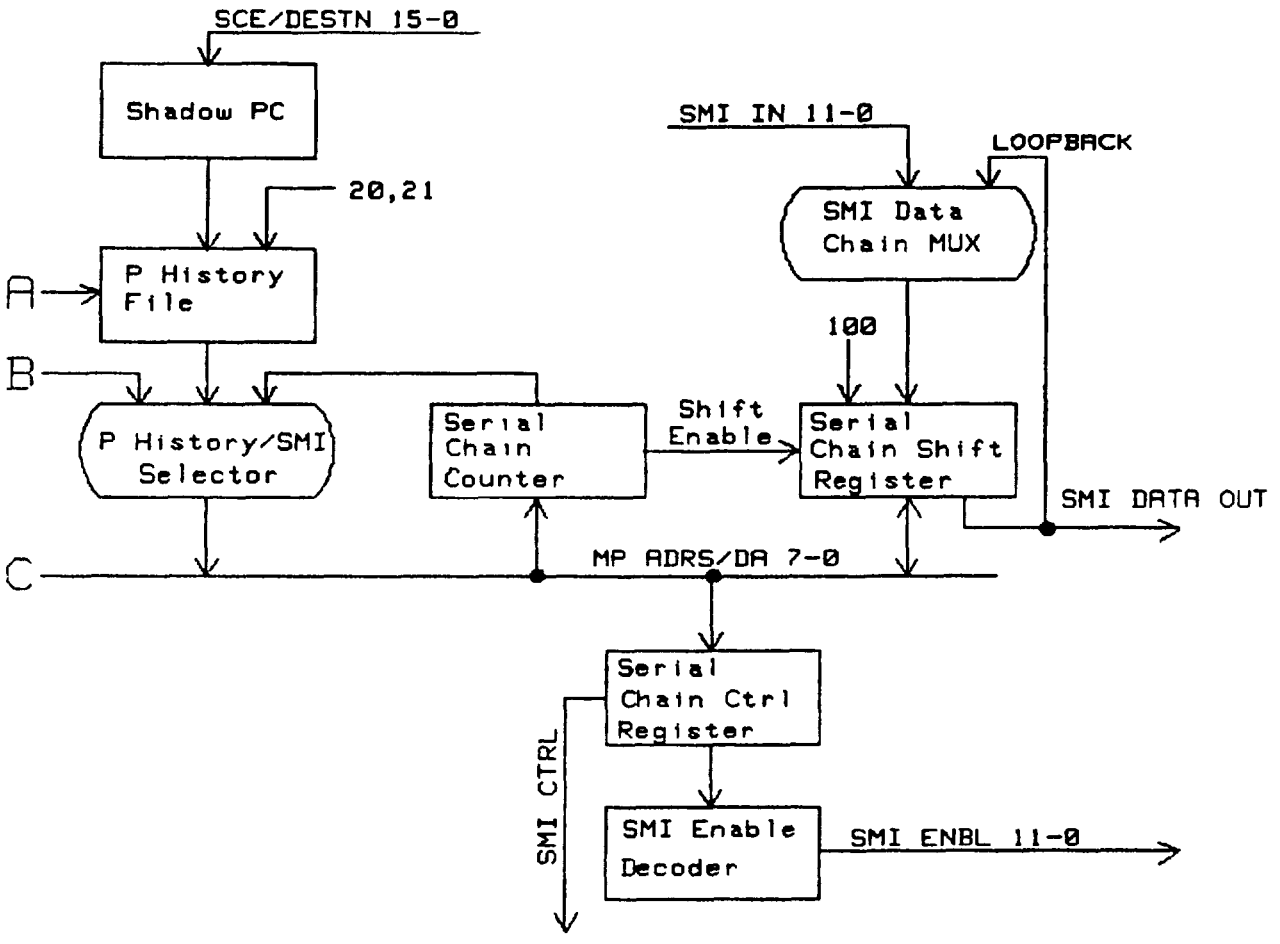


FIGURE 1. CWH functional block diagram - Continued.

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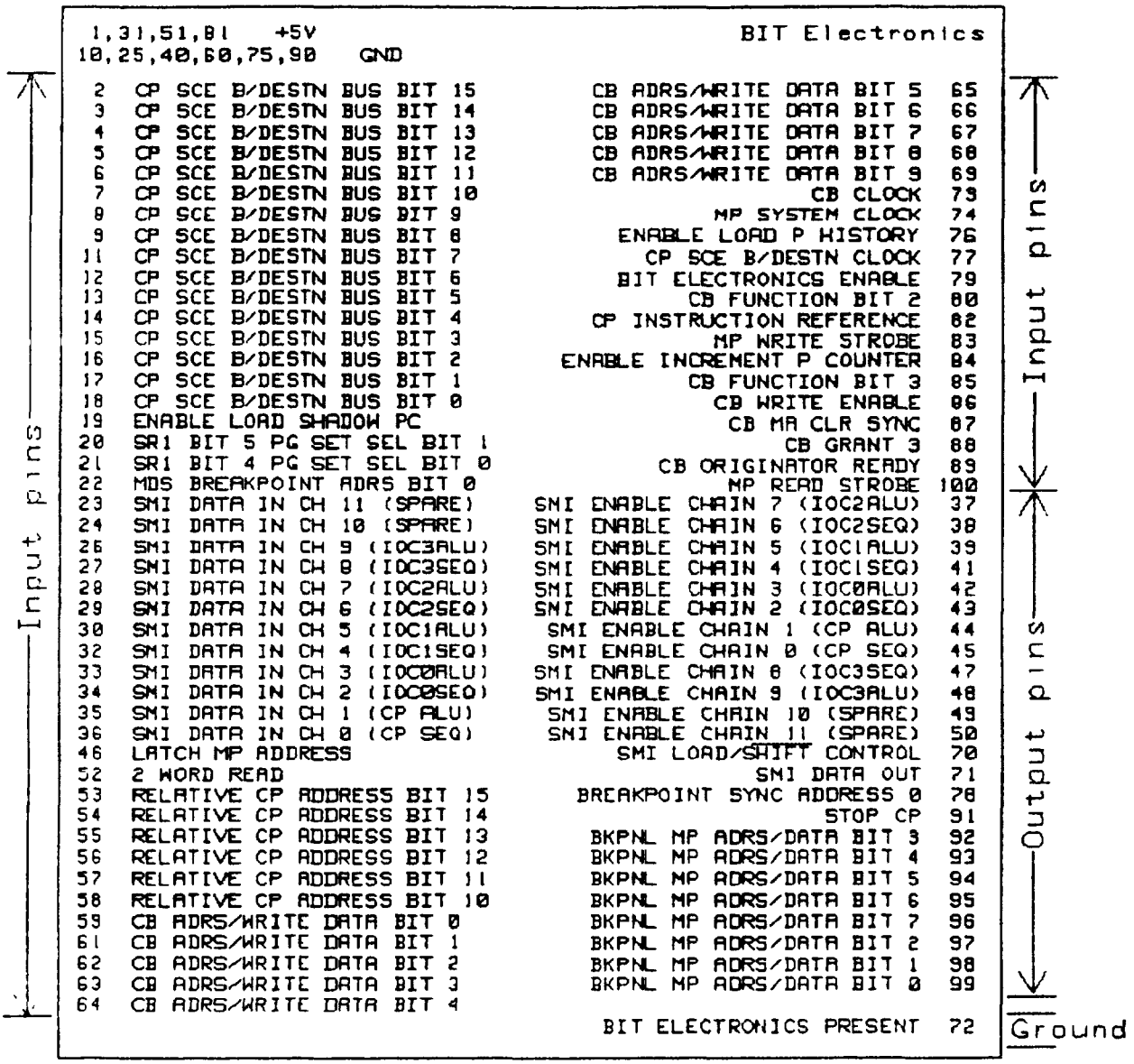


FIGURE 2. CWH pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

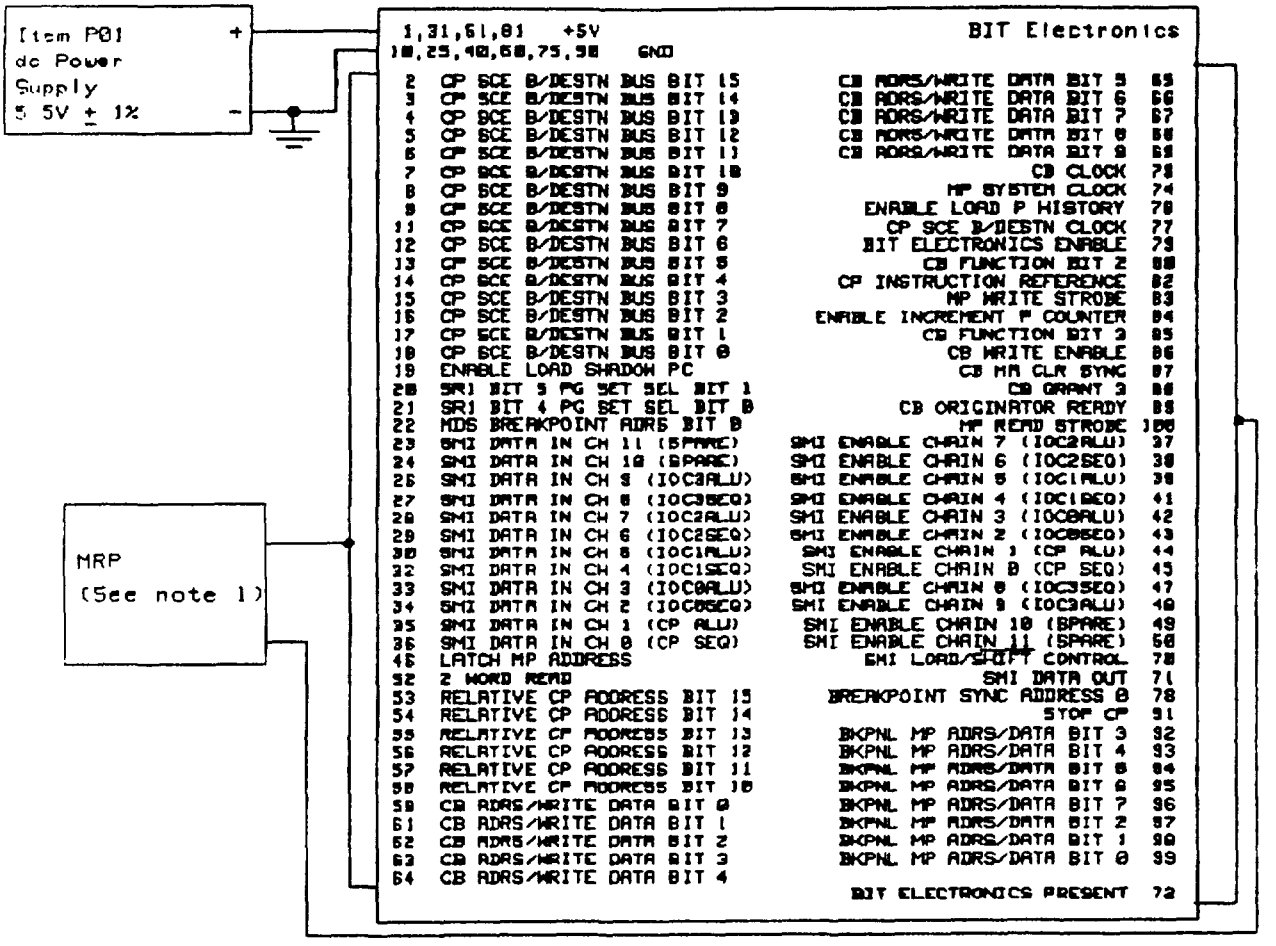
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. CWH operating temperature and component temperature test circuit.

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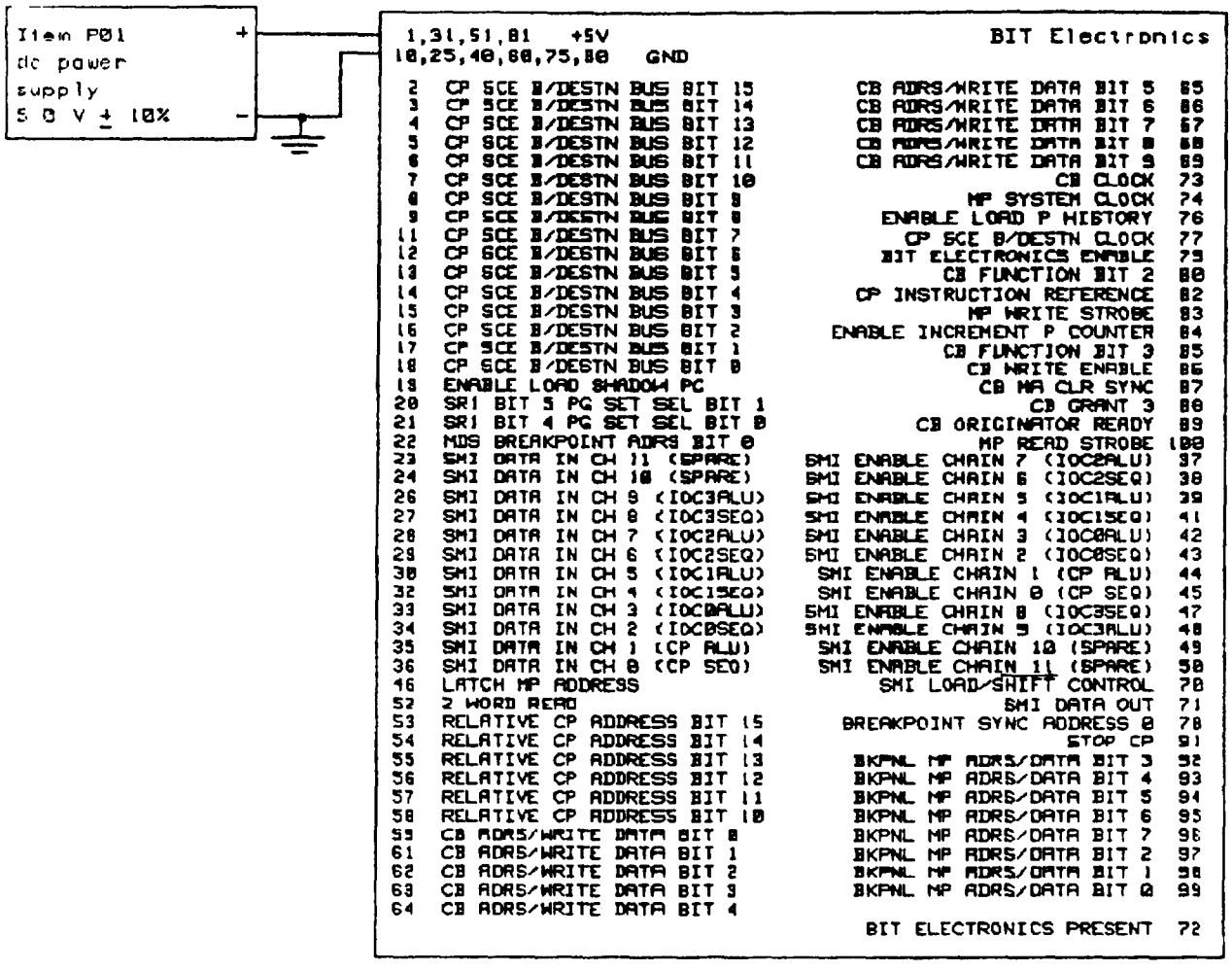


FIGURE 4. CWH vibration test circuit.

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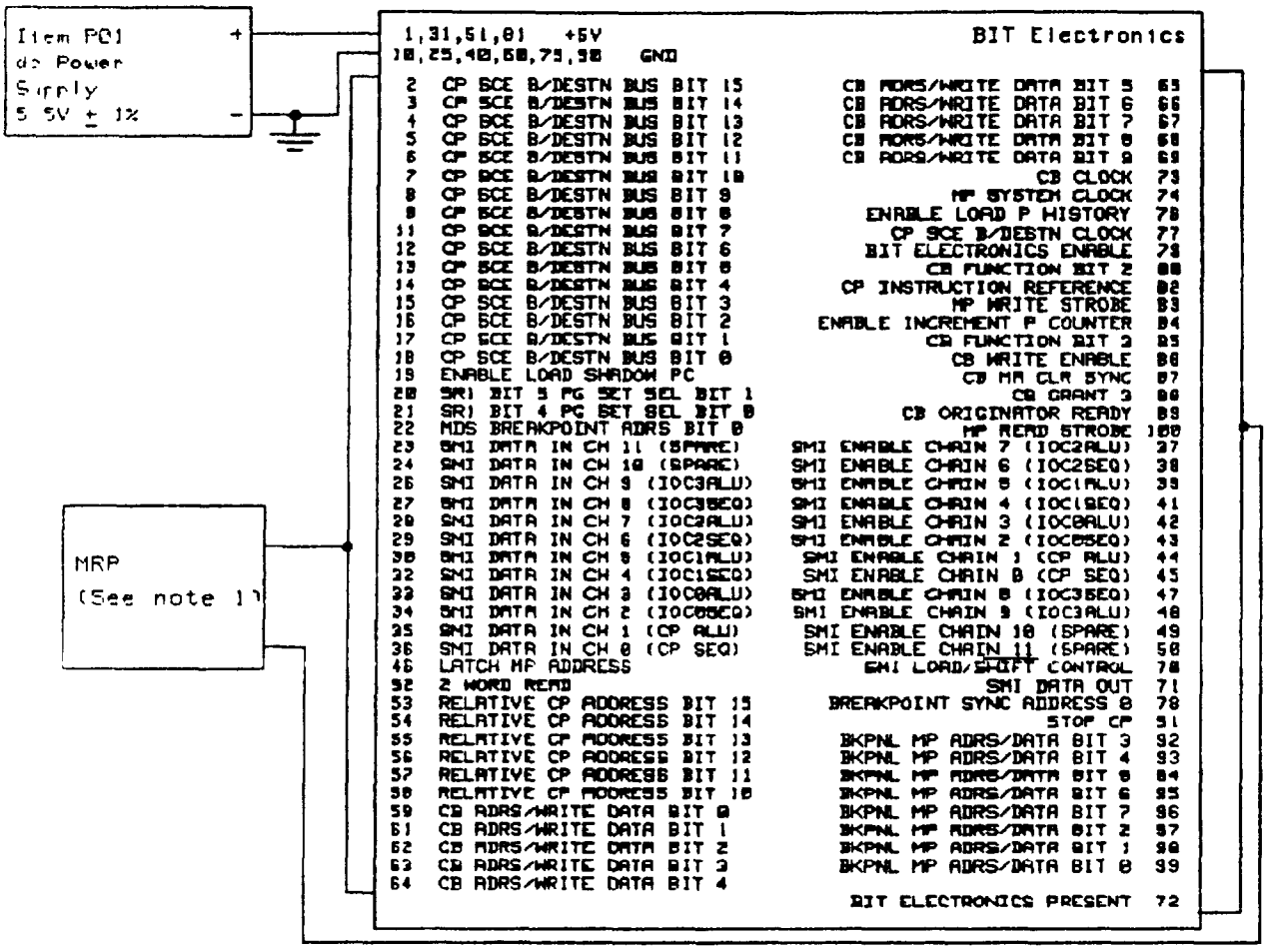
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. CWH life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) NA = not applicable.
(l) PSFC = power supply filter capacitance.
(m) PSTA = power supply transient amplitude.
(n) Vic = input clamp diode voltage.
(o) Vih = input high level voltage.
(p) Vil = input low level voltage.
(q) Vin = input forcing voltage.
(r) Voh = output high level voltage.
(s) Vol = output low level voltage.
(t) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 6.14 W (max); 4.12 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE/DESTN 15	-1.5	0.4	2.4	-0.4	0.02	0.1	50
SCE/DESTN 14	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 13	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 12	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 11	-1.5	0.4	2.4	-0.4	0.02	0.1	50
SCE/DESTN 10	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 9	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 8	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 7	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 6	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 5	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 4	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 3	-1.5	0.4	2.4	-0.4	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SCE/DESTN 2	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SCE/DESTN 0	-1.5	0.4	2.4	-0.4	0.02	0.1	75
ENBL PC	-1.5	0.4	2.4	-1.2	0.08	0.4	100
PG SET/SEL 1	NA	0.4	2.4	-0.8	0.01	1.0	75
PG SET/SEL 0	NA	0.4	2.4	-0.8	0.01	1.0	75
MDS ADRS 0	NA	0.4	2.4	-1.75	-0.59	0.12	150
SMI IN 11	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 10	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 9	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 8	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 7	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 6	-1.5	0.4	2.4	-0.4	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SMI IN 5	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 4	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 3	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 2	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
SMI IN 0	-1.5	0.4	2.4	-0.4	0.02	0.1	75
MP ADDR	-1.5	0.4	2.4	-0.2	0.02	0.1	75
2 WORD READ	NA	0.4	2.4	-2.35	-0.57	0.22	100
CP ADDR 15	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CP ADDR 14	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CP ADDR 13	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CP ADDR 12	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CP ADDR 11	-1.5	0.4	2.4	-0.2	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CP ADDR 10	-1.5	0.4	2.4	-0.2	0.02	0.1	75
CB ADRS/WR 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB ADRS/WR 1	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 2	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 3	-1.5	0.4	2.4	-0.36	0.02	0.1	50
CB ADRS/WR 4	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 5	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 6	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 7	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 8	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB ADRS/WR 9	-1.5	0.4	2.4	-0.36	0.02	0.1	75
CB CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	75
MP CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ENBL PH	-1.2	0.4	2.4	-3.2	0.08	2.1	150
CP CLK	-1.2	0.4	2.4	-6.0	0.15	3.0	100
BIT ENBL	-1.5	0.4	2.4	-0.6	0.04	0.2	100
CB FUNC 2	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CP INSTR REF	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MP WR STROBE	-1.5	0.4	2.4	-0.4	0.02	0.1	75
EN INC P CNT	NA	0.4	2.4	-1.75	-0.59	0.12	75
CB FUNC 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB WR ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MACLR SYNC	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB GRANT 3	-1.2	0.4	2.4	-2.4	0.04	0.2	100
CB ORIG RDY	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MP RD STROBE	-1.2	0.4	2.4	-1.4	0.06	0.3	100

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Ios max (mA)
SMI ENBL 7	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 6	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 5	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 4	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 3	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 2	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 1	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 0	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 8	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 9	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 10	2.4	0.5	-0.4	6.0	NA	-60.0
SMI ENBL 11	2.4	0.5	-0.4	6.0	NA	-60.0
SMI CTRL	2.4	0.5	-0.4	6.0	NA	-100.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (V)	Iol max (mA)	Iozh max (mA)	Ios max (mA)
SMI DATA OUT	2.4	0.5	-0.4	6.0	NA	-100.0
SYNC ADRS 0	2.4	0.5	-0.4	6.0	NA	-60.0
STOP CP	NA	0.5	NA	6.0	0.25	NA
MP ADRS/DA 3	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 4	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 5	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 6	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 7	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 2	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 1	2.4	0.5	-0.16	1.6	NA	-40.0
MP ADRS/DA 0	2.4	0.5	-0.16	1.6	NA	-40.0

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	SCE/DESTN 0 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 1 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 2 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 3 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 4 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 5 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 6 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 7 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 8 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 9 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 10 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 11 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 12 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 13 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 14 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 15 to CP CLK	32	32	32	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 0	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 1	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 2	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 3	15	15	15	-	-	-	nS	12a

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	th	CP CLK to SCE/DESTN 4	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 5	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 6	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 7	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 8	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 9	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 10	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 11	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 12	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 13	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 14	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 15	15	15	15	-	-	-	nS	12a
3	tsu	PG SET/SEL 1 to CP CLK	52	52	52	-	-	-	nS	12a
3	tsu	PG SET/SEL 0 to CP CLK	52	52	52	-	-	-	nS	12a
4	th	CP CLK to PG SET/SEL 1	10	10	10	-	-	-	nS	12a
4	th	CP CLK to PG SET/SEL 0	10	10	10	-	-	-	nS	12a
5	tsu	ENBL PC to CP CLK	42	42	42	-	-	-	nS	12a

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MP CLK to SMI ENBL 7	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 6	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 5	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 4	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 3	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 2	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 1	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 0	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 8	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 9	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 10	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 11	-	-	-	50	50	50	nS	12b
7	tpd	MP WR STROBE to SMI CTRL	-	-	-	73	73	73	nS	12b
8	tpd	MP CLK to SMI DATA OUT	-	-	-	41	41	41	nS	12b
9	tpd	CB CLK to STOP CP	-	-	-	41	41	41	nS	12b

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6.4 Module function. The BIT electronics module contains the Serial Maintenance Interface (SMI), breakpoint function, and Program (P) history file. The SMI provides a serial data path between the processor and the Central Processor (CP) sequencer and Arithmetic Logic Unit (ALU) modules and Input/Output Controller (IOC) sequencer and IOC ALU modules. The processor tests and verifies correct operation of the module by reading and/or writing a serial chain of data/status information (self test, errors, parity errors, timeouts, tristate bus enables, etc.). In addition, the serial chain for the sequencer module includes the microinstruction register and micro start/stop flip-flops that allow the processor to load and execute CP and IOC microinstructions. The breakpoint function, when enabled, stops the MRP when the referenced memory address matches the address in the breakpoint register. The breakpoint can be enabled for a read, write, or instruction memory reference (all or any combination can be enabled). The P history file is used to record the address and source of the last eight events that caused a branch from the next sequential instruction.

6.4.1 P history circuits. The P history circuits consist of the P history file, shadow program counter, P history counter, and P history address selector. The P history circuits record the address and source of the last eight events that caused a branch from the next sequential instruction. The shadow Program Counter (PC) contains the address of the instruction currently being executed. (The shadow PC contains same address as PC). When a branch condition occurs, the address in shadow PC is loaded into the P history file along with the event that caused the branch. The address is contained in P history file bits 15-0 and the event in bits 19-16 as follows:

(a) Bit 19 - Load status register 1.

(b) Bit 18 - Load page register.

(c) Bit 17 - Jump instruction.

(d) Bit 16 - Interrupt.

A 20-bit register within the file is addressed via the P history selector from either the P history counter or Micro Processor (MP) address latch. Breakpoint/P history control register bit 3 determines the selection (bit 3 = 0, select P history counter; bit 3 = 1, select MP address latch bits 7-4). The P history

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counter contains the number of branch conditions that occur and is used to address a P history file register to record the event. The counter is incremented by one for each branch condition. The counter is cleared by a master clear or when breakpoint/P history control register bit 6 equals one. MP address bits 7-4 address the P history file for maintenance processor read operations. The maintenance processor addresses the selector to read a P history file register and the P history counter as follows:

(a) Address 7E01 - P history bits 7-0.

(b) Address 7E02 - P history bits 15-8.

(c) Address 7E03 - P history counter 3-0 (address/data bits 7-4) and P history bits 19-16 (address/data bits 3-0).

6.4.2 Control circuits. The control circuits consist of a decoder and miscellaneous circuits. The control circuits translate processor addresses in the 7EXX range to provide the enables necessary to transfer data on the BIT electronics module.

6.4.3 MP address latch. The latch holds the processor address bits 7-0. The latch is enabled by the address latch enable signal from the processor.

6.4.4 SMI data chain multiplexer. The multiplexer inputs the selected data chain to the serial chain shift register. Serial chain control register bits 3-0 select the data chain.

6.4.5 Serial chain shift register. The shift register assembles incoming serial data into an 8-bit byte to transfer to the processor or receives an 8-bit byte from the processor, converts it to serial data, and transfers it to the selected module. The processor addresses the shift register as 7E06 to load or read the register contents. The register is right shifted at the sync clock rate when receiving or sending serial data. Shifting is controlled by the serial chain counter. The counter contains the number of bits to shift. When the count reaches zero, shifting stops.

6.4.6 Serial chain counter. The counter holds the count that determines the number of times the data in the serial chain shift register is right shifted. The processor addresses the counter as 7E05 to load the bit count. This also enables the counter which decrements at the sync clock rate. The processor monitors the counter by reading its contents via the selector with an address of 7E00.

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6.4.7 Serial chain control register. The 8-bit control register holds the address of the active chain in bits 3-0 and the load/shift control in bit 7 (bit 7 = 0, load; bit 7 = 1, shift). The load/shift bit is sent to the selected module. The processor addresses the register as 7E03 to load.

6.4.8 SMI enable decoder. The decoder translates serial chain control register bits 3-0 to enable the SMI chain with the selected module. The decoder is enabled when the serial chain counter is loaded and remains enabled until the counter reaches zero.

6.4.9 Breakpoint/P history control register. The 8-bit control register holds breakpoint and P history control information and provides other control functions. The processor addresses the register as 7E02 to load the control word.

6.4.10 Function comparator. The function comparator compares the current memory operation (read operand, write operand, or instruction fetch) with the corresponding control bit (read breakpoint, write breakpoint, or instruction breakpoint) in the breakpoint control register. When they compare, the breakpoint comparator is enabled to compare the address in the breakpoint register with the memory address.

6.4.11 Breakpoint register. The 16-bit register holds the breakpoint address. The processor addresses the lower byte as 7E00 and the upper byte as 7E01 to load the register.

6.4.12 Address register. The 6-bit register holds relative address bits 15-10. These bits along with common bus bits 9-0 form the memory address.

6.4.13 Breakpoint comparator. The breakpoint comparator, when enabled by the function comparator, compares the address in the breakpoint register with the memory address (address register bits 15-10 and common bus bits 9-0). When they compare, the stop signal is generated to stop program execution. The stop is cleared when breakpoint control register bit 6 is set.

6.4.14 Address/Data bus transceiver. The transceiver provides two-way communication between the BIT electronics module bus and the maintenance processor module bus. The transceiver is enabled by an address in the 7EXX range. Data transfer direction is determined by the read signal from the processor.

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6.4.15 Description of I/O pins.

Signal name	Pin number	I/O	Description
SCE/DESTN 15	2	I	CP SCE B/DESTN BUS BIT 15
SCE/DESTN 14	3	I	CP SCE B/DESTN BUS BIT 14
SCE/DESTN 13	4	I	CP SCE B/DESTN BUS BIT 13
SCE/DESTN 12	5	I	CP SCE B/DESTN BUS BIT 12
SCE/DESTN 11	6	I	CP SCE B/DESTN BUS BIT 11
SCE/DESTN 10	7	I	CP SCE B/DESTN BUS BIT 10
SCE/DESTN 9	8	I	CP SCE B/DESTN BUS BIT 9
SCE/DESTN 8	9	I	CP SCE B/DESTN BUS BIT 8
SCE/DESTN 7	11	I	CP SCE B/DESTN BUS BIT 7
SCE/DESTN 6	12	I	CP SCE B/DESTN BUS BIT 6
SCE/DESTN 5	13	I	CP SCE B/DESTN BUS BIT 5
SCE/DESTN 4	14	I	CP SCE B/DESTN BUS BIT 4
SCE/DESTN 3	15	I	CP SCE B/DESTN BUS BIT 3
SCE/DESTN 2	16	I	CP SCE B/DESTN BUS BIT 2
SCE/DESTN 1	17	I	CP SCE B/DESTN BUS BIT 1
SCE/DESTN 0	18	I	CP SCE B/DESTN BUS BIT 0
ENBL PC	19	I	ENABLE LOAD SHADOW PC
PG SET/SEL 1	20	I	SR1 BIT 5 PG SET SEL BIT 1
PG SET/SEL 0	21	I	SR1 BIT 4 PG SET SEL BIT 0
MDS ADRS 0	22	I	MDS BREAKPOINT ADRS BIT 0

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6.4.15 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SMI IN 11	23	I	SMI DATA IN CH 11 (SPARE)
SMI IN 10	24	I	SMI DATA IN CH 10 (SPARE)
SMI IN 9	26	I	SMI DATA IN CH 9 (IOC3ALU)
SMI IN 8	27	I	SMI DATA IN CH 8 (IOC3SEQ)
SMI IN 7	28	I	SMI DATA IN CH 7 (IOC2ALU)
SMI IN 6	29	I	SMI DATA IN CH 6 (IOC2SEQ)
SMI IN 5	30	I	SMI DATA IN CH 5 (IOC1ALU)
SMI IN 4	32	I	SMI DATA IN CH 4 (IOC1SEQ)
SMI IN 3	33	I	SMI DATA IN CH 3 (IOC0ALU)
SMI IN 2	34	I	SMI DATA IN CH 2 (IOC0SEQ)
SMI IN 1	35	I	SMI DATA IN CH 1 (CP ALU)
SMI IN 0	36	I	SMI DATA IN CH 0 (CP SEQ)
MP ADDR	46	I	LATCH MP ADDRESS
2 WORD READ	52	I	2 WORD READ
CP ADDR 15	53	I	RELATIVE CP ADDRESS BIT 15
CP ADDR 14	54	I	RELATIVE CP ADDRESS BIT 14
CP ADDR 13	55	I	RELATIVE CP ADDRESS BIT 13
CP ADDR 12	56	I	RELATIVE CP ADDRESS BIT 12
CP ADDR 11	57	I	RELATIVE CP ADDRESS BIT 11
CP ADDR 10	58	I	RELATIVE CP ADDRESS BIT 10

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6.4.15 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB ADRS/WR 0	59	I	CB ADRS/WRITE DATA BIT 0
CB ADRS/WR 1	61	I	CB ADRS/WRITE DATA BIT 1
CB ADRS/WR 2	62	I	CB ADRS/WRITE DATA BIT 2
CB ADRS/WR 3	63	I	CB ADRS/WRITE DATA BIT 3
CB ADRS/WR 4	64	I	CB ADRS/WRITE DATA BIT 4
CB ADRS/WR 5	65	I	CB ADRS/WRITE DATA BIT 5
CB ADRS/WR 6	66	I	CB ADRS/WRITE DATA BIT 6
CB ADRS/WR 7	67	I	CB ADRS/WRITE DATA BIT 7
CB ADRS/WR 8	68	I	CB ADRS/WRITE DATA BIT 8
CB ADRS/WR 9	69	I	CB ADRS/WRITE DATA BIT 9
CB CLK	73	I	CB CLOCK
MP CLK	74	I	MP SYSTEM CLOCK
ENBL PH	76	I	ENABLE LOAD P HISTORY
CP CLK	77	I	CP SCE B/DESTN CLOCK
BIT ENBL	79	I	BIT ELECTRONICS ENABLE
CB FUNC 2	80	I	CB FUNCTION BIT 2
CP INSTR REF	82	I	CP INSTRUCTION REFERENCE
MP WR STROBE	83	I	MP WRITE STROBE
EN INC P CNT	84	I	ENABLE INCREMENT P COUNTER
CB FUNC 3	85	I	CB FUNCTION BIT 3

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6.4.15 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB WR ENBL	86	I	CB WRITE ENABLE
MACLR SYNC	87	I	CB MA CLR SYNC
CB GRANT 3	88	I	CB GRANT 3
CB ORIG RDY	89	I	CB ORIGINATOR READY
MP RD STROBE	100	I	MP READ STROBE
SMI ENBL 7	37	O	SMI ENABLE CHAIN 7 (IOC2ALU)
SMI ENBL 6	38	O	SMI ENABLE CHAIN 6 (IOC2SEQ)
SMI ENBL 5	39	O	SMI ENABLE CHAIN 5 (IOC1ALU)
SMI ENBL 4	41	O	SMI ENABLE CHAIN 4 (IOC1SEQ)
SMI ENBL 3	42	O	SMI ENABLE CHAIN 3 (IOC0ALU)
SMI ENBL 2	43	O	SMI ENABLE CHAIN 2 (IOC0SEQ)
SMI ENBL 1	44	O	SMI ENABLE CHAIN 1 (CP ALU)
SMI ENBL 0	45	O	SMI ENABLE CHAIN 0 (CP SEQ)
SMI ENBL 8	47	O	SMI ENABLE CHAIN 8 (IOC3SEQ)
SMI ENBL 9	48	O	SMI ENABLE CHAIN 9 (IOC3ALU)
SMI ENBL 10	49	O	SMI ENABLE CHAIN 10 (SPARE)
SMI ENBL 11	50	O	SMI ENABLE CHAIN 11 (SPARE)
SMI CTRL	70	O	SMI LOAD/$\overline{\text{SHIFT}}$ CONTROL
SMI DATA OUT	71	O	SMI DATA OUT
SYNC ADRS 0	78	O	BREAKPOINT SYNC ADDRESS 0

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6.4.15 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
STOP CP	91	O	STOP CP
MP ADRS/DA 3	92	O	BKPNL MP ADRS/DATA BIT 3
MP ADRS/DA 4	93	O	BKPNL MP ADRS/DATA BIT 4
MP ADRS/DA 5	94	O	BKPNL MP ADRS/DATA BIT 5
MP ADRS/DA 6	95	O	BKPNL MP ADRS/DATA BIT 6
MP ADRS/DA 7	96	O	BKPNL MP ADRS/DATA BIT 7
MP ADRS/DA 2	97	O	BKPNL MP ADRS/DATA BIT 2
MP ADRS/DA 1	98	O	BKPNL MP ADRS/DATA BIT 1
MP ADRS/DA 0	99	O	BKPNL MP ADRS/DATA BIT 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N736)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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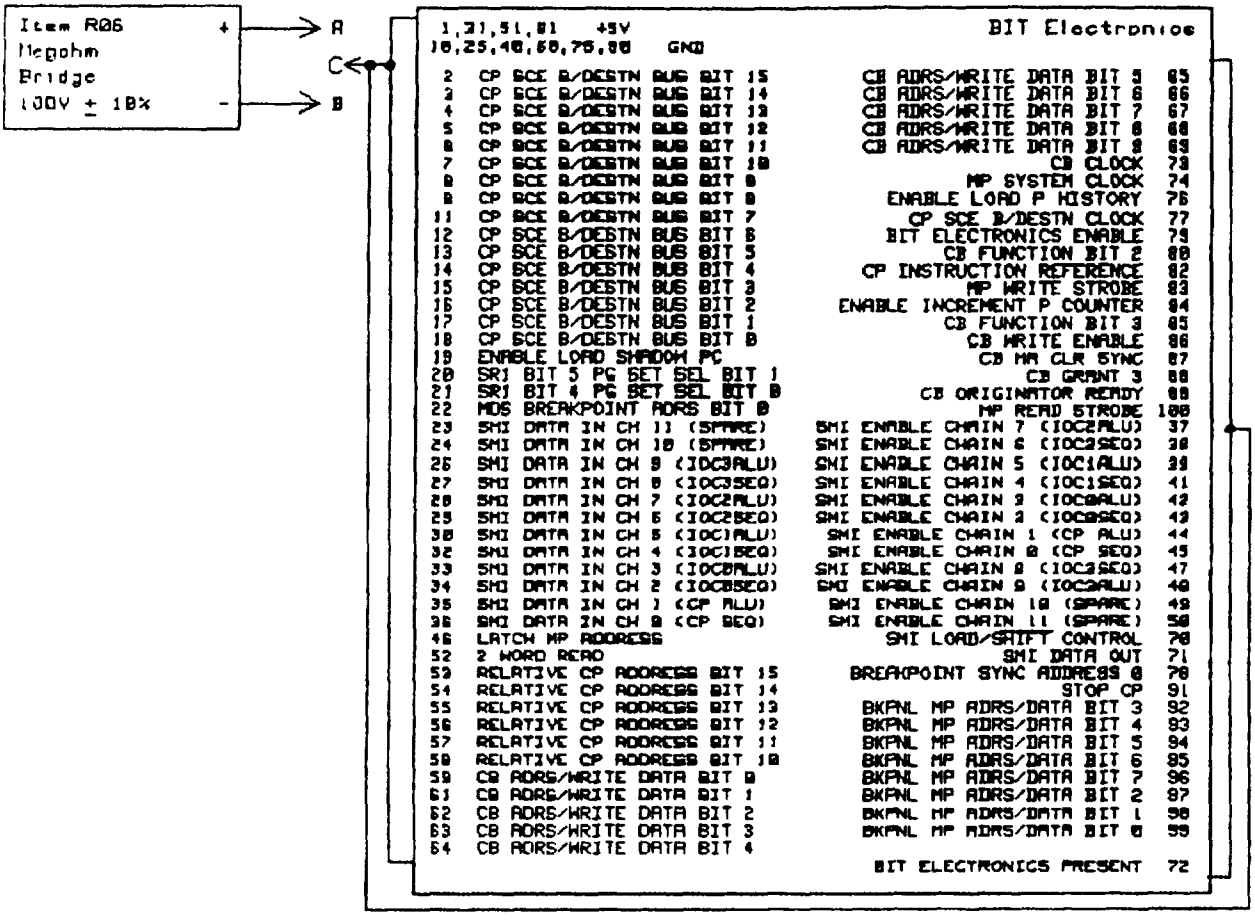


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81				
B7	Continuity, GND	60	72	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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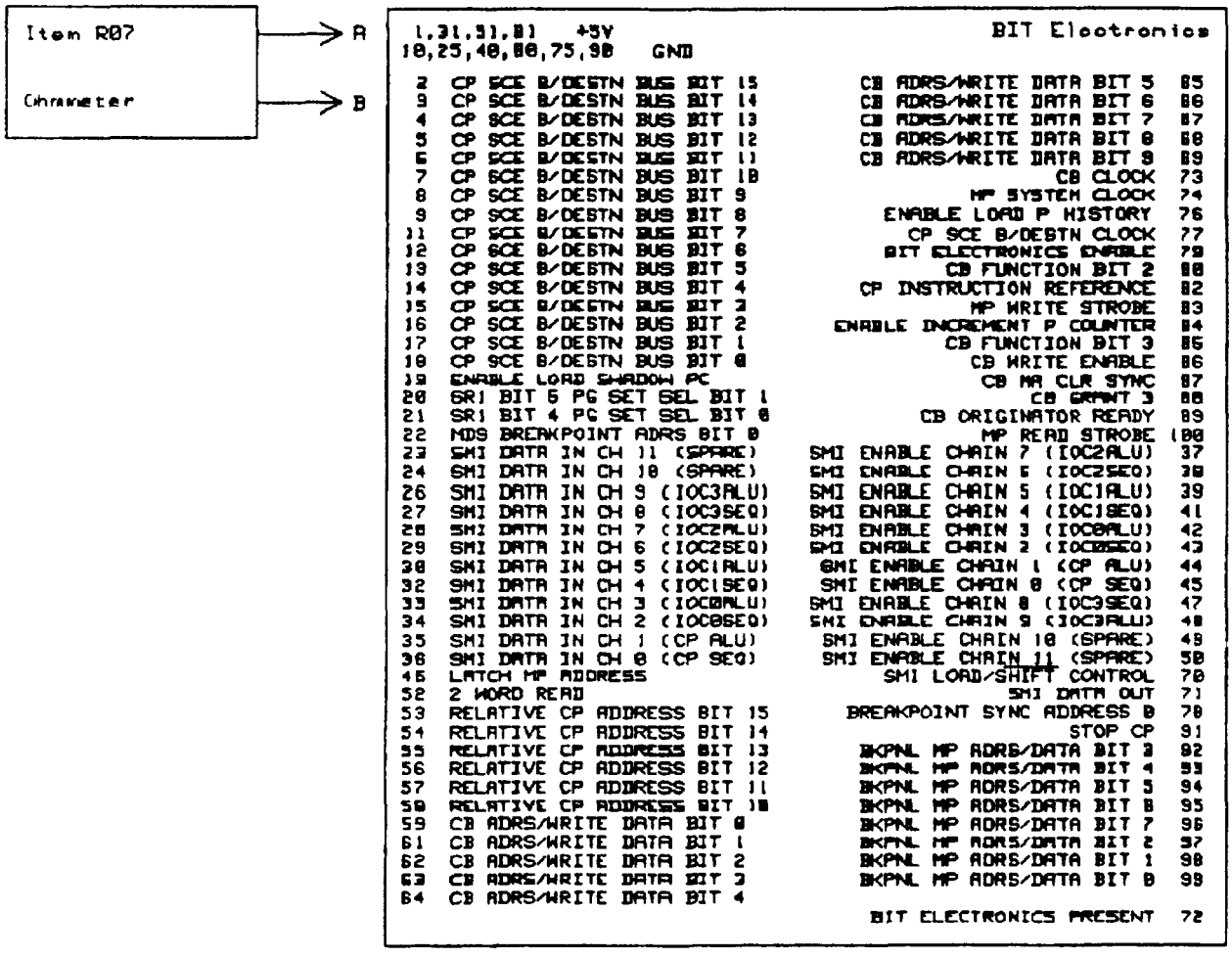


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all re-maining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.01 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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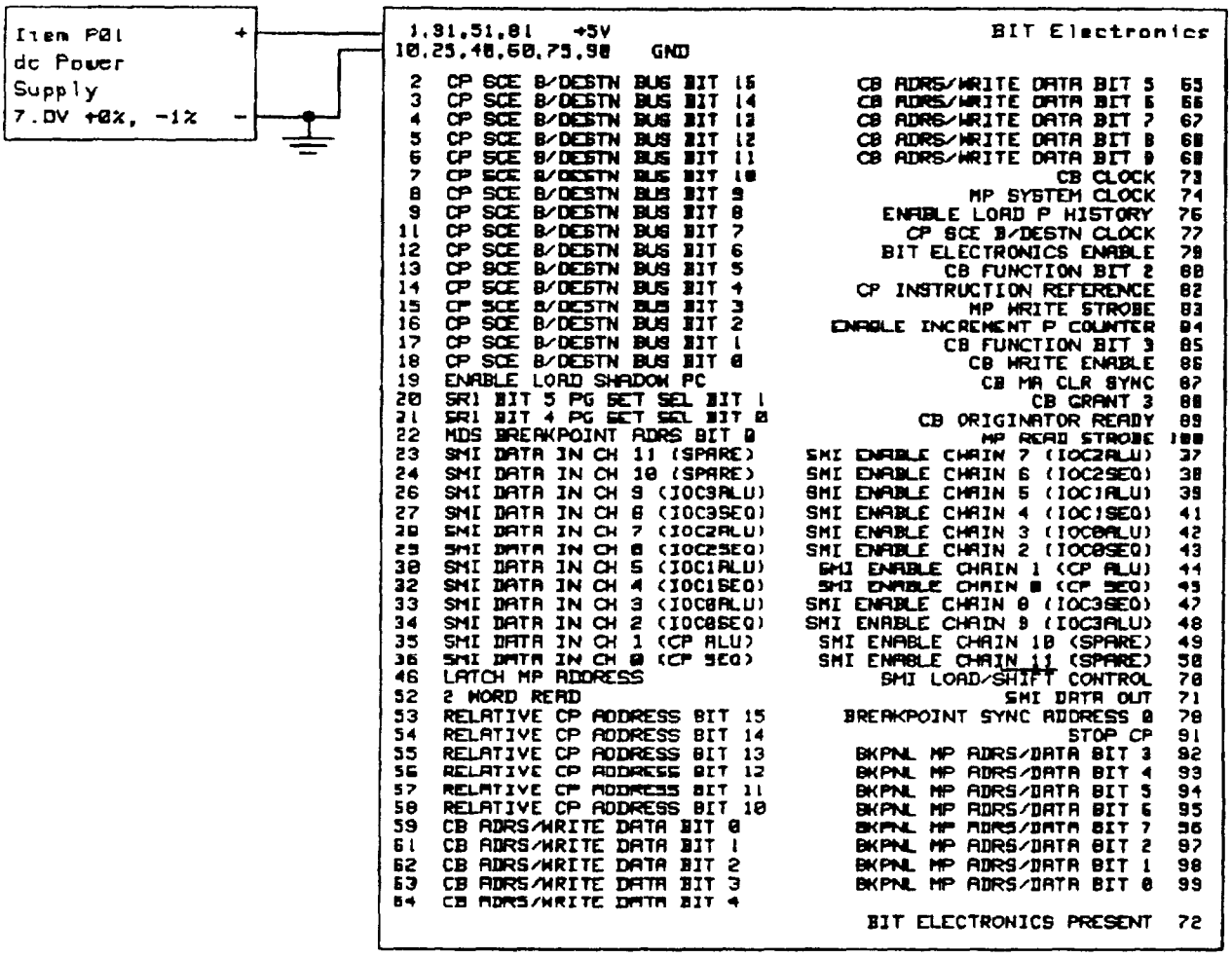


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	SCE/DESTN 15	2	5.5	0.0	0.1	mA	1-3
D2	SCE/DESTN 14	3					
D3	SCE/DESTN 13	4					
D4	SCE/DESTN 12	5					
D5	SCE/DESTN 11	6					
D6	SCE/DESTN 10	7					
D7	SCE/DESTN 9	8					
D8	SCE/DESTN 8	9					
D9	SCE/DESTN 7	11					
D10	SCE/DESTN 6	12					
D11	SCE/DESTN 5	13					
D12	SCE/DESTN 4	14					
D13	SCE/DESTN 3	15					
D14	SCE/DESTN 2	16					
D15	SCE/DESTN 1	17					
D16	SCE/DESTN 0	18	5.5	0.0	0.1	mA	1-3
D17	ENBL PC	19	5.5	0.0	0.4	mA	1-3
D18	PG SET/SEL 1	20	5.5	0.0	1.0	mA	1-3
D19	PG SET/SEL 0	21	5.5	0.0	1.0	mA	1-3
D20	MDS ADRS 0	22	5.5	-0.02	0.12	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	SMI IN 11	23	5.5	0.0	0.1	mA	1-3
D22	SMI IN 10	24					
D23	SMI IN 9	26					
D24	SMI IN 8	27					
D25	SMI IN 7	28					
D26	SMI IN 6	29					
D27	SMI IN 5	30					
D28	SMI IN 4	32					
D29	SMI IN 3	33					
D30	SMI IN 2	34					
D31	SMI IN 1	35					
D32	SMI IN 0	36					
D33	MP ADDR	46	5.5	0.0	0.1	mA	1-3
D34	2 WORD READ	52	5.5	-0.02	0.22	mA	1-3
D35	CP ADDR 15	53	5.5	0.0	0.1	mA	1-3
D36	CP ADDR 14	54					
D37	CP ADDR 13	55					
D38	CP ADDR 12	56					
D39	CP ADDR 11	57					
D40	CP ADDR 10	58	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	CB ADRS/WR 0	59	5.5	0.0	0.1	mA	1-3
D42	CB ADRS/WR 1	61					
D43	CB ADRS/WR 2	62					
D44	CB ADRS/WR 3	63					
D45	CB ADRS/WR 4	64					
D46	CB ADRS/WR 5	65					
D47	CB ADRS/WR 6	66					
D48	CB ADRS/WR 7	67					
D49	CB ADRS/WR 8	68					
D50	CB ADRS/WR 9	69	5.5	0.0	0.1	mA	1-3
D51	CB CLK	73	5.5	0.0	1.0	mA	1-3
D52	MP CLK	74	5.5	0.0	1.0	mA	1-3
D53	ENBL PH	76	5.5	0.0	2.1	mA	1-3
D54	CP CLK	77	5.5	0.0	3.0	mA	1-3
D55	BIT ENBL	79	5.5	0.0	0.2	mA	1-3
D56	CB FUNC 2	80					
D57	CP INSTR REF	82	5.5	0.0	0.2	mA	1-3
D58	MP WR STROBE	83	5.5	0.0	0.1	mA	1-3
D59	EN INC P CNT	84	5.5	-0.02	0.12	mA	1-3
D60	CB FUNC 3	85	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D61	CB WR ENBL	86	5.5	0.0	0.1	mA	1-3
D62	MACLR SYNC	87	5.5	0.0	0.1	mA	1-3
D63	CB GRANT 3	88	5.5	0.0	0.2	mA	1-3
D64	CB ORIG RDY	89	5.5	0.0	0.1	mA	1-3
D65	MP RD STROBE	100	5.5	0.0	0.3	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D65
Power supply current limit	1.67 A	NA	D1-D65
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D65
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D65
low, (VL)	0.0 V	± 100 mV	D1-D65
Input current measurement	NA	± 5 uA	D1-D16
	NA	± 20 uA	D17
	NA	± 50 uA	D18,D19
	NA	± 7 uA	D20
	NA	± 5 uA	D21-D33
	NA	± 12 uA	D34
	NA	± 5 uA	D35-D50
	NA	± 50 uA	D51,D52
	NA	± 105 uA	D53
	NA	± 150 uA	D54
	NA	± 10 uA	D55-D57
	NA	± 5 uA	D58
	NA	± 7 uA	D59
	NA	± 5 uA	D60-D62
	NA	± 10 uA	D63
	NA	± 5 uA	D64
	NA	± 15 uA	D65

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	SMI ENBL 7	37	0.0	-150.0	-60.0	mA	1-4
E2	SMI ENBL 6	38					
E3	SMI ENBL 5	39					
E4	SMI ENBL 4	41					
E5	SMI ENBL 3	42					
E6	SMI ENBL 2	43					
E7	SMI ENBL 1	44					
E8	SMI ENBL 0	45					
E9	SMI ENBL 8	47					
E10	SMI ENBL 9	48					
E11	SMI ENBL 10	49					
E12	SMI ENBL 11	50	0.0	-150.0	-60.0	mA	1-4
E13	SMI CTRL	70	0.0	-325.0	-100.0	mA	1-4
E14	SMI DATA OUT	71	0.0	-325.0	-100.0	mA	1-4
E15	SYNC ADRS 0	78	0.0	-150.0	-60.0	mA	1-4
E16	MP ADRS/DA 3	92	0.0	-225.0	-40.0	mA	1-4
E17	MP ADRS/DA 4	93					
E18	MP ADRS/DA 5	94					
E19	MP ADRS/DA 6	95					
E20	MP ADRS/DA 7	96	0.0	-225.0	-40.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	MP ADRS/DA 2	97	0.0	-225.0	-40.0	mA	1-4
E22	MP ADRS/DA 1	98	|	|	|	|	|
E23	MP ADRS/DA 0	99	0.0	-225.0	-40.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E23
Power supply current limit	1.67 A	NA	E1-E23
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E23
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E23 E1-E23
Output current measurement	NA NA NA NA	± 4.5 mA ± 11.2 mA ± 4.5 mA ± 9.25 mA	E1-E12 E13,E14 E15 E16-E23

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	SCE/DESTN 15	2	2.4	-0.4	0.02	mA	1-3
F2	SCE/DESTN 14	3					
F3	SCE/DESTN 13	4					
F4	SCE/DESTN 12	5					
F5	SCE/DESTN 11	6					
F6	SCE/DESTN 10	7					
F7	SCE/DESTN 9	8					
F8	SCE/DESTN 8	9					
F9	SCE/DESTN 7	11					
F10	SCE/DESTN 6	12					
F11	SCE/DESTN 5	13					
F12	SCE/DESTN 4	14					
F13	SCE/DESTN 3	15					
F14	SCE/DESTN 2	16					
F15	SCE/DESTN 1	17					
F16	SCE/DESTN 0	18	2.4	-0.4	0.02	mA	1-3
F17	ENBL PC	19	2.4	-1.2	0.08	mA	1-3
F18	PG SET/SEL 1	20	2.4	-0.8	0.01	mA	1-3
F19	PG SET/SEL 0	21	2.4	-0.8	0.01	mA	1-3
F20	MDS ADRS 0	22	2.4	-1.75	-0.59	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	SMI IN 11	23	2.4	-0.4	0.02	mA	1-3
F22	SMI IN 10	24					
F23	SMI IN 9	26					
F24	SMI IN 8	27					
F25	SMI IN 7	28					
F26	SMI IN 6	29					
F27	SMI IN 5	30					
F28	SMI IN 4	32					
F29	SMI IN 3	33					
F30	SMI IN 2	34					
F31	SMI IN 1	35					
F32	SMI IN 0	36	2.4	-0.4	0.02	mA	1-3
F33	MP ADDR	46	2.4	-0.2	0.02	mA	1-3
F34	2 WORD READ	52	2.4	-2.35	-0.57	mA	1-3
F35	CP ADDR 15	53	2.4	-0.2	0.02	mA	1-3
F36	CP ADDR 14	54					
F37	CP ADDR 13	55					
F38	CP ADDR 12	56					
F39	CP ADDR 11	57					
F40	CP ADDR 10	58	2.4	-0.2	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	CB ADRS/WR 0	59	2.4	-0.6	0.02	mA	1-3
F42	CB ADRS/WR 1	61	2.4	-0.36	0.02	mA	1-3
F43	CB ADRS/WR 2	62					
F44	CB ADRS/WR 3	63					
F45	CB ADRS/WR 4	64					
F46	CB ADRS/WR 5	65					
F47	CB ADRS/WR 6	66					
F48	CB ADRS/WR 7	67					
F49	CB ADRS/WR 8	68					
F50	CB ADRS/WR 9	69	2.4	-0.36	0.02	mA	1-3
F51	CB CLK	73	2.4	-2.0	0.05	mA	1-3
F52	MP CLK	74	2.4	-2.0	0.05	mA	1-3
F53	ENBL PH	76	2.4	-3.2	0.08	mA	1-3
F54	CP CLK	77	2.4	-6.0	0.15	mA	1-3
F55	BIT ENBL	79	2.4	-0.6	0.04	mA	1-3
F56	CB FUNC 2	80	2.4	-1.2	0.04	mA	1-3
F57	CP INSTR REF	82	2.4	-1.2	0.04	mA	1-3
F58	MP WR STROBE	83	2.4	-0.4	0.02	mA	1-3
F59	EN INC P CNT	84	2.4	-1.75	-0.59	mA	1-3
F60	CB FUNC 3	85	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	CB WR ENBL	86	2.4	-0.6	0.02	mA	1-3
F62	MACLR SYNC	87	2.4	-0.6	0.02	mA	1-3
F63	CB GRANT 3	88	2.4	-2.4	0.04	mA	1-3
F64	CB ORIG RDY	89	2.4	-0.6	0.02	mA	1-3
F65	MP RD STROBE	100	2.4	-1.4	0.06	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:29Z
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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F65
Power supply current limit	1.67 A	NA	F1-F65
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F65
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F65
low, (VL)	0.0 V	± 100 mV	F1-F65
Input current measurement	NA	± 21.0 uA	F1-F16
	NA	± 64.0 uA	F17
	NA	± 40.5 uA	F18,F19
	NA	± 58.0 uA	F20
	NA	± 21.0 uA	F21-F32
	NA	± 11.0 uA	F33
	NA	± 89.0 uA	F34
	NA	± 11.0 uA	F35-F40
	NA	± 31.0 uA	F41
	NA	± 19.0 uA	F42-F50
	NA	± 102.0 uA	F51,F52
	NA	± 164.0 uA	F53
	NA	± 307.0 uA	F54
	NA	± 32.0 uA	F55
	NA	± 62.0 uA	F56,F57
	NA	± 21.0 uA	F58
	NA	± 58.0 uA	F59
	NA	± 31.0 uA	F60-F62
	NA	± 122.0 uA	F63
	NA	± 31.0 uA	F64
	NA	± 73.0 uA	F65

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	SCE/DESTN 15	2	0.4	-0.4	0.02	mA	1-3
G2	SCE/DESTN 14	3					
G3	SCE/DESTN 13	4					
G4	SCE/DESTN 12	5					
G5	SCE/DESTN 11	6					
G6	SCE/DESTN 10	7					
G7	SCE/DESTN 9	8					
G8	SCE/DESTN 8	9					
G9	SCE/DESTN 7	11					
G10	SCE/DESTN 6	12					
G11	SCE/DESTN 5	13					
G12	SCE/DESTN 4	14					
G13	SCE/DESTN 3	15					
G14	SCE/DESTN 2	16					
G15	SCE/DESTN 1	17					
G16	SCE/DESTN 0	18	0.4	-0.4	0.02	mA	1-3
G17	ENBL PC	19	0.4	-1.2	0.08	mA	1-3
G18	PG SET/SEL 1	20	0.4	-0.8	0.01	mA	1-3
G19	PG SET/SEL 0	21	0.4	-0.8	0.01	mA	1-3
G20	MDS ADRS 0	22	0.4	-1.75	-0.59	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G21	SMI IN 11	23	0.4	-0.4	0.02	mA	1-3
G22	SMI IN 10	24					
G23	SMI IN 9	26					
G24	SMI IN 8	27					
G25	SMI IN 7	28					
G26	SMI IN 6	29					
G27	SMI IN 5	30					
G28	SMI IN 4	32					
G29	SMI IN 3	33					
G30	SMI IN 2	34					
G31	SMI IN 1	35					
G32	SMI IN 0	36	0.4	-0.4	0.02	mA	1-3
G33	MP ADDR	46	0.4	-0.2	0.02	mA	1-3
G34	2 WORD READ	52	0.4	-2.35	-0.57	mA	1-3
G35	CP ADDR 15	53	0.4	-0.2	0.02	mA	1-3
G36	CP ADDR 14	54					
G37	CP ADDR 13	55					
G38	CP ADDR 12	56					
G39	CP ADDR 11	57					
G40	CP ADDR 10	58	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	CB ADRS/WR 0	59	0.4	-0.6	0.02	mA	1-3
G42	CB ADRS/WR 1	61	0.4	-0.36	0.02	mA	1-3
G43	CB ADRS/WR 2	62					
G44	CB ADRS/WR 3	63					
G45	CB ADRS/WR 4	64					
G46	CB ADRS/WR 5	65					
G47	CB ADRS/WR 6	66					
G48	CB ADRS/WR 7	67					
G49	CB ADRS/WR 8	68					
G50	CB ADRS/WR 9	69	0.4	-0.36	0.02	mA	1-3
G51	CB CLK	73	0.4	-2.0	0.05	mA	1-3
G52	MP CLK	74	0.4	-2.0	0.05	mA	1-3
G53	ENBL PH	76	0.4	-3.2	0.08	mA	1-3
G54	CP CLK	77	0.4	-6.0	0.15	mA	1-3
G55	BIT ENBL	79	0.4	-0.6	0.04	mA	1-3
G56	CB FUNC 2	80	0.4	-1.2	0.04	mA	1-3
G57	CP INSTR REF	82	0.4	-1.2	0.04	mA	1-3
G58	MP WR STROBE	83	0.4	-0.4	0.02	mA	1-3
G59	EN INC P CNT	84	0.4	-1.75	-0.59	mA	1-3
G60	CB FUNC 3	85	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	CB WR ENBL	86	0.4	-0.6	0.02	mA	1-3
G62	MACLR SYNC	87	0.4	-0.6	0.02	mA	1-3
G63	CB GRANT 3	88	0.4	-2.4	0.04	mA	1-3
G64	CB ORIG RDY	89	0.4	-0.6	0.02	mA	1-3
G65	MP RD STROBE	100	0.4	-1.4	0.06	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G65
Power supply current limit	1.67 A	NA	G1-G65
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G65
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G65
low, (VL)	0.0 V	± 100 mV	G1-G65
Input current measurement	NA	± 21.0 uA	G1-G16
	NA	± 64.0 uA	G17
	NA	± 40.5 uA	G18,G19
	NA	± 58.0 uA	G20
	NA	± 21.0 uA	G21-G32
	NA	± 11.0 uA	G33
	NA	± 89.0 uA	G34
	NA	± 11.0 uA	G35-G40
	NA	± 31.0 uA	G41
	NA	± 19.0 uA	G42-G50
	NA	± 102.0 uA	G51,G52
	NA	± 164.0 uA	G53
	NA	± 307.0 uA	G54
	NA	± 32.0 uA	G55
	NA	± 62.0 uA	G56,G57
	NA	± 21.0 uA	G58
	NA	± 58.0 uA	G59
	NA	± 31.0 uA	G60-G62
	NA	± 122.0 uA	G63
	NA	± 31.0 uA	G64
	NA	± 73.0 uA	G65

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	SMI ENBL 7	37	-0.4	2.4	4.5	V	1-4
H2	SMI ENBL 6	38					
H3	SMI ENBL 5	39					
H4	SMI ENBL 4	41					
H5	SMI ENBL 3	42					
H6	SMI ENBL 2	43					
H7	SMI ENBL 1	44					
H8	SMI ENBL 0	45					
H9	SMI ENBL 8	47					
H10	SMI ENBL 9	48					
H11	SMI ENBL 10	49					
H12	SMI ENBL 11	50					
H13	SMI CTRL	70					
H14	SMI DATA OUT	71					
H15	SYNC ADRS 0	78	-0.4	2.4	4.5	V	1-4
H16	MP ADRS/DA 3	92	-0.16	2.4	4.5	V	1-4
H17	MP ADRS/DA 4	93					
H18	MP ADRS/DA 5	94					
H19	MP ADRS/DA 6	95					
H20	MP ADRS/DA 7	96	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	MP ADRS/DA 2	97	-0.16	2.4	4.5	V	1-4
H22	MP ADRS/DA 1	98	↓	↓	↓	↓	↓
H23	MP ADRS/DA 0	99	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H23
Power supply current limit	1.67 A	NA	H1-H23
Forcing current, (Ioh)	-0.4 mA -0.16 mA	± 40 uA ± 16 uA	H1-H15 H16-H23
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H23 H1-H23
Output voltage measurement	NA	± 105 mV	H1-H23

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

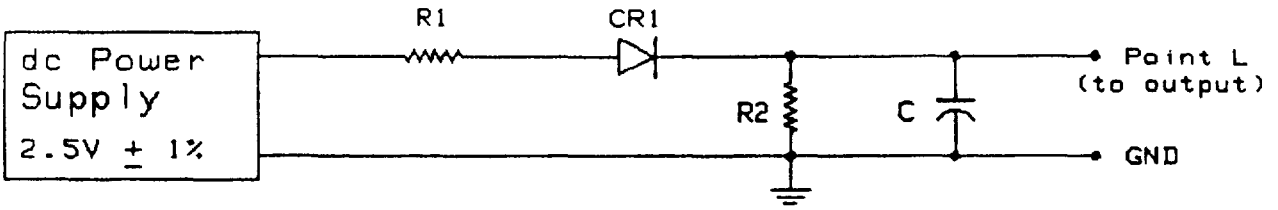
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
37,38,39,41,42, 43,44,45,47,48, 49,50,70,71,78	6200	6.0	0.5
91	NA	6.0	0.5
92,93,94,95,96, 97,98,99	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	SMI ENBL 7	37	6.0	0.0	0.5	V	1-4
J2	SMI ENBL 6	38					
J3	SMI ENBL 5	39					
J4	SMI ENBL 4	41					
J5	SMI ENBL 3	42					
J6	SMI ENBL 2	43					
J7	SMI ENBL 1	44					
J8	SMI ENBL 0	45					
J9	SMI ENBL 8	47					
J10	SMI ENBL 9	48					
J11	SMI ENBL 10	49					
J12	SMI ENBL 11	50					
J13	SMI CTRL	70					
J14	SMI DATA OUT	71					
J15	SYNC ADRS 0	78					
J16	STOP CP	91	6.0	0.0	0.5	V	1-4
J17	MP ADRS/DA 3	92	1.6	0.0	0.5	V	1-4
J18	MP ADRS/DA 4	93					
J19	MP ADRS/DA 5	94					
J20	MP ADRS/DA 6	95	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	MP ADRS/DA 7	96	1.6	0.0	0.5	V	1-4
J22	MP ADRS/DA 2	97					
J23	MP ADRS/DA 1	98					
J24	MP ADRS/DA 0	99	1.6	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:29Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J24
Power supply current limit	1.67 A	NA	J1-J24
Forcing current, (Iol)	6.0 mA 1.6 mA	± 600 uA ± 160 uA	J1-J16 J17-J24
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J24 J1-J24
Output voltage measurement	NA	± 25 mV	J1-J24

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K	STOP CP	91	5.5	0.0	0.25	mA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K
Power supply current limit	1.67 A	NA	K
Forcing voltage, (Voh)	5.5 V	± 50 mV	K
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K K
Output current measurement	NA	± 12.5 uA	K

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	SCE/DESTN 15	2	-18.0	-1.5	0.0	V	1-3
L2	SCE/DESTN 14	3					
L3	SCE/DESTN 13	4					
L4	SCE/DESTN 12	5					
L5	SCE/DESTN 11	6					
L6	SCE/DESTN 10	7					
L7	SCE/DESTN 9	8					
L8	SCE/DESTN 8	9					
L9	SCE/DESTN 7	11					
L10	SCE/DESTN 6	12					
L11	SCE/DESTN 5	13					
L12	SCE/DESTN 4	14					
L13	SCE/DESTN 3	15					
L14	SCE/DESTN 2	16					
L15	SCE/DESTN 1	17					
L16	SCE/DESTN 0	18					
L17	ENBL PC	19					
L18	SMI IN 11	23					
L19	SMI IN 10	24					
L20	SMI IN 9	26	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	SMI IN 8	27	-18.0	-1.5	0.0	V	1-3
L22	SMI IN 7	28					
L23	SMI IN 6	29					
L24	SMI IN 5	30					
L25	SMI IN 4	32					
L26	SMI IN 3	33					
L27	SMI IN 2	34					
L28	SMI IN 1	35					
L29	SMI IN 0	36					
L30	MP ADDR	46					
L31	CP ADDR 15	53					
L32	CP ADDR 14	54					
L33	CP ADDR 13	55					
L34	CP ADDR 12	56					
L35	CP ADDR 11	57					
L36	CP ADDR 10	58	-18.0	-1.5	0.0	V	1-3
L37	CB ADRS/WR 0	59	-18.0	-1.2	0.0	V	1-3
L38	CB ADRS/WR 1	61	-18.0	-1.5	0.0	V	1-3
L39	CB ADRS/WR 2	62					
L40	CB ADRS/WR 3	63	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L41	CB ADRS/WR 4	64	-18.0	-1.5	0.0	V	1-3
L42	CB ADRS/WR 5	65					
L43	CB ADRS/WR 6	66					
L44	CB ADRS/WR 7	67					
L45	CB ADRS/WR 8	68					
L46	CB ADRS/WR 9	69	-18.0	-1.5	0.0	V	1-3
L47	CB CLK	73	-18.0	-1.2	0.0	V	1-3
L48	MP CLK	74					
L49	ENBL PH	76					
L50	CP CLK	77	-18.0	-1.2	0.0	V	1-3
L51	BIT ENBL	79	-18.0	-1.5	0.0	V	1-3
L52	CB FUNC 2	80	-18.0	-1.2	0.0	V	1-3
L53	CP INSTR REF	82	-18.0	-1.2	0.0	V	1-3
L54	MP WR STROBE	83	-18.0	-1.5	0.0	V	1-3
L55	CB FUNC 3	85	-18.0	-1.2	0.0	V	1-3
L56	CB WR ENBL	86					
L57	MACLR SYNC	87					
L58	CB GRANT 3	88					
L59	CB ORIG RDY	89					
L60	MP RD STROBE	100	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L60
Power supply current limit	NA	NA	L1-L60
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L60
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L60 L1-L60
Input voltage measurement	NA NA NA NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1-L36 L37 L38-L46 L47-L50 L51 L52,L53 L54 L55-L60

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	7	13	uF	1-4
M2	PSFC	51	6	12	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 300 nF ± 300 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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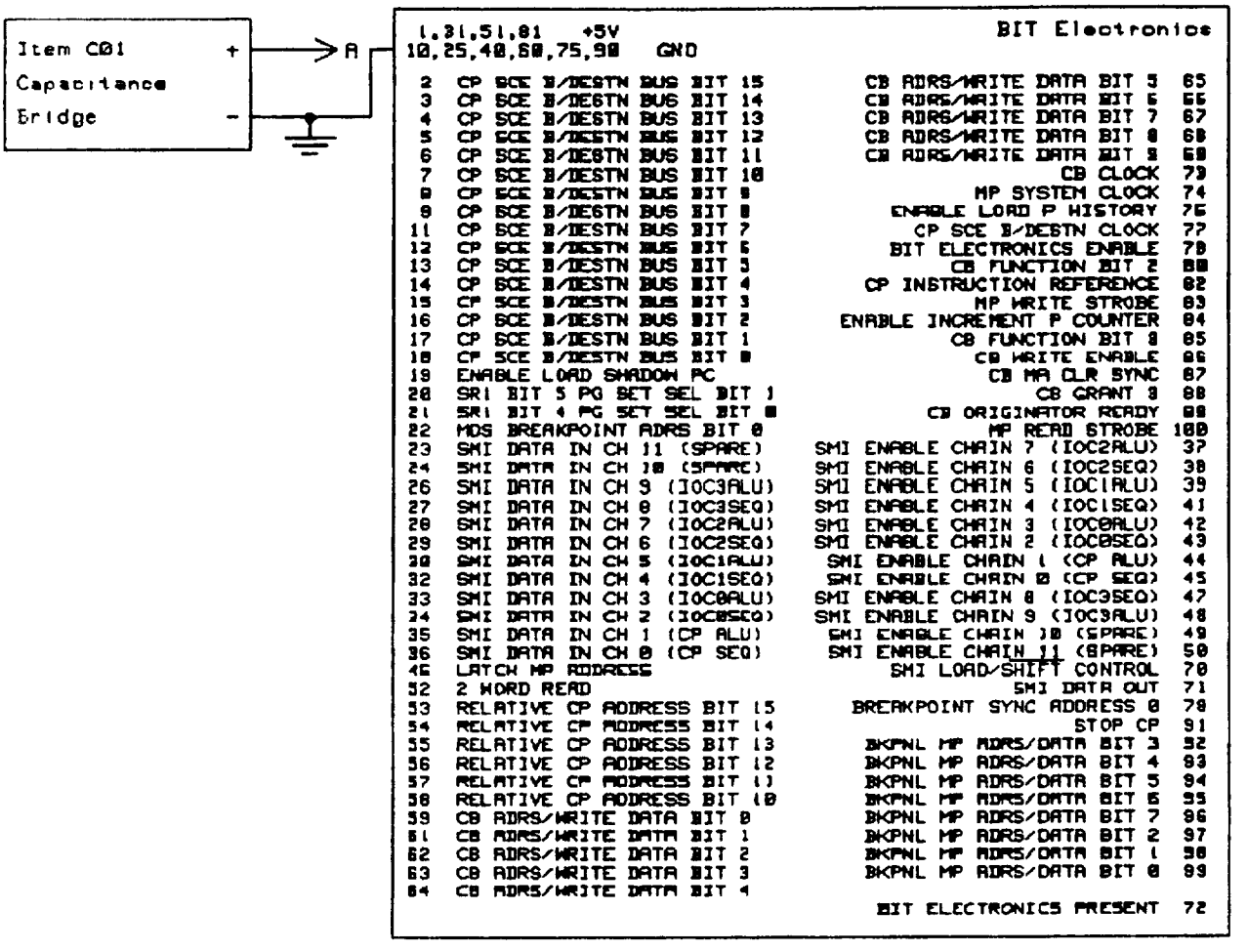


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	465	1116	mA	1-3

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NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.67 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 32.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.67 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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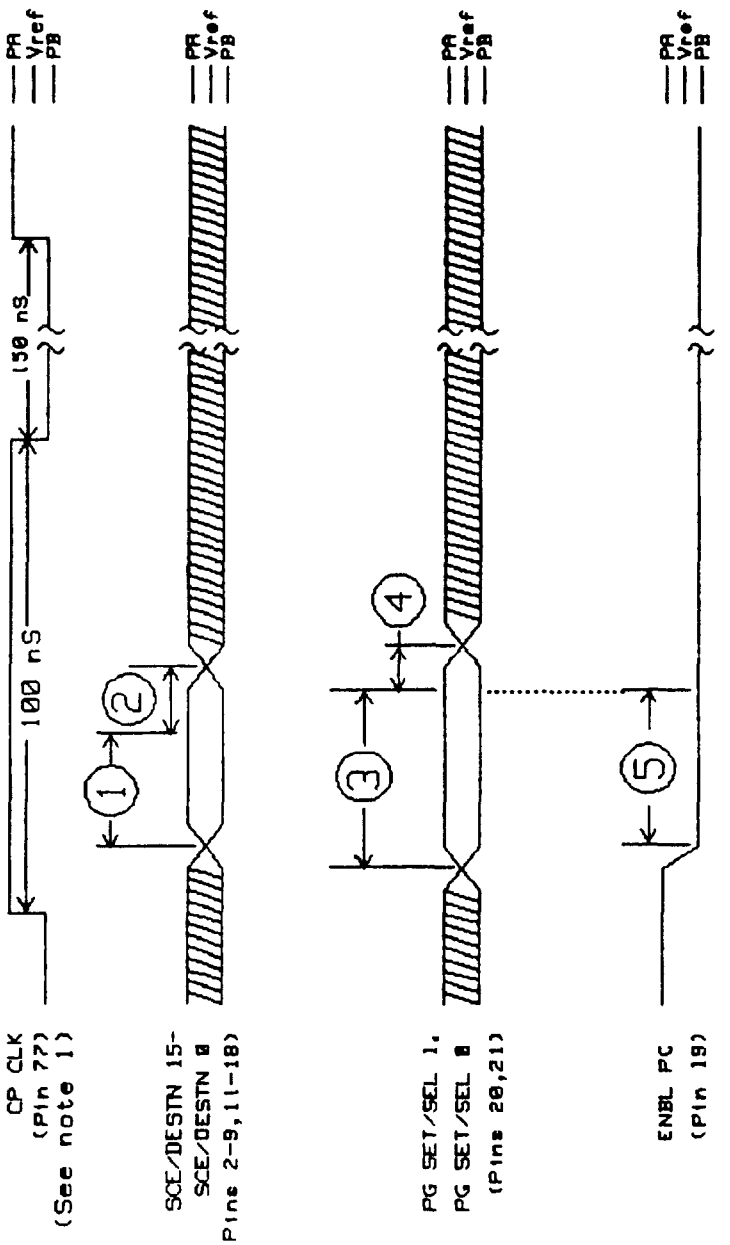


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. The waveform is not drawn to proportion.

2. (1) SCE/DESTN 15 - SCE/DESTN 0 to CP CLK. Setup time: 32 nS minimum.

3. (2) CP CLK to SCE/DESTN 15 - SCE/DESTN 0. Hold time: 15 nS minimum.

4. (3) PG SET/SEL 1 and PG SET/SEL 0 to CP CLK. Setup time: 52 nS minimum.

5. (4) CP CLK to PG SET/SEL 1 and PG SET/SEL 0. Hold time: 10 nS minimum.

6. (5) ENBL PC to CP CLK. Setup time: 42 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.67 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

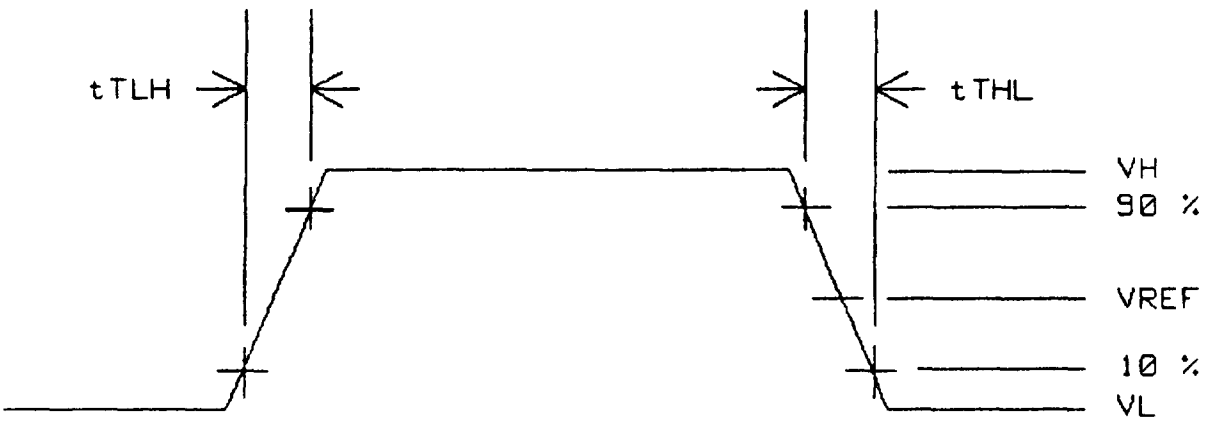
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref Num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	SCE/DESTN 0 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 1 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 2 to CP CLK	32	32	32	-	-	-	nS	12a
	tsu	SCE/DESTN 3 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 4 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 5 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 6 to CP CLK	32	32	32	-	-	-	nS	12a
	tsu	SCE/DESTN 7 to CP CLK	32	32	32	-	-	-	nS	12a
	tsu	SCE/DESTN 8 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 9 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 10 to CP CLK	32	32	32	-	-	-	nS	12a
	tsu	SCE/DESTN 11 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 12 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 13 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 14 to CP CLK	32	32	32	-	-	-	nS	12a
1	tsu	SCE/DESTN 15 to CP CLK	32	32	32	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 0	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 1	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 2	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 3	15	15	15	-	-	-	nS	12a

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TABLE XVII. Dynamic parameters - Continued.

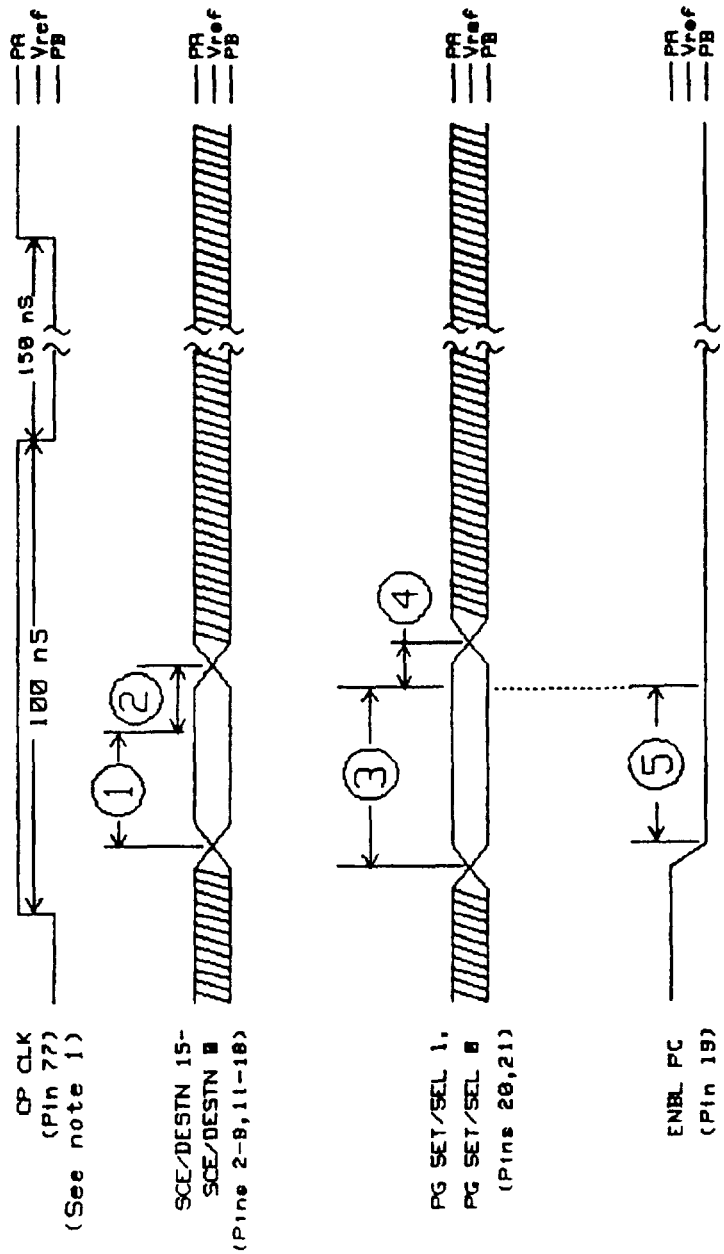
Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	th	CP CLK to SCE/DESTN 4	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 5	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 6	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 7	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 8	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 9	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 10	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 11	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 12	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 13	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 14	15	15	15	-	-	-	nS	12a
2	th	CP CLK to SCE/DESTN 15	15	15	15	-	-	-	nS	12a
3	tsu	PG SET/SEL 1 to CP CLK	52	52	52	-	-	-	nS	12a
3	tsu	PG SET/SEL 0 to CP CLK	52	52	52	-	-	-	nS	12a
4	th	CP CLK to PG SET/SEL 1	10	10	10	-	-	-	nS	12a
4	th	CP CLK to PG SET/SEL 0	10	10	10	-	-	-	nS	12a
5	tsu	ENBL PC to CP CLK	42	42	42	-	-	-	nS	12a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MP CLK to SMI ENBL 7	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 6	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 5	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 4	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 3	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 2	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 1	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 0	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 8	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 9	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 10	-	-	-	50	50	50	nS	12b
6	tpd	MP CLK to SMI ENBL 11	-	-	-	50	50	50	nS	12b
7	tpd	MP WR STROBE to SMI CTRL	-	-	-	73	73	73	nS	12b
8	tpd	MP CLK to SMI DATA OUT	-	-	-	41	41	41	nS	12b
9	tpd	CB CLK to STOP CP	-	-	-	41	41	41	nS	12b

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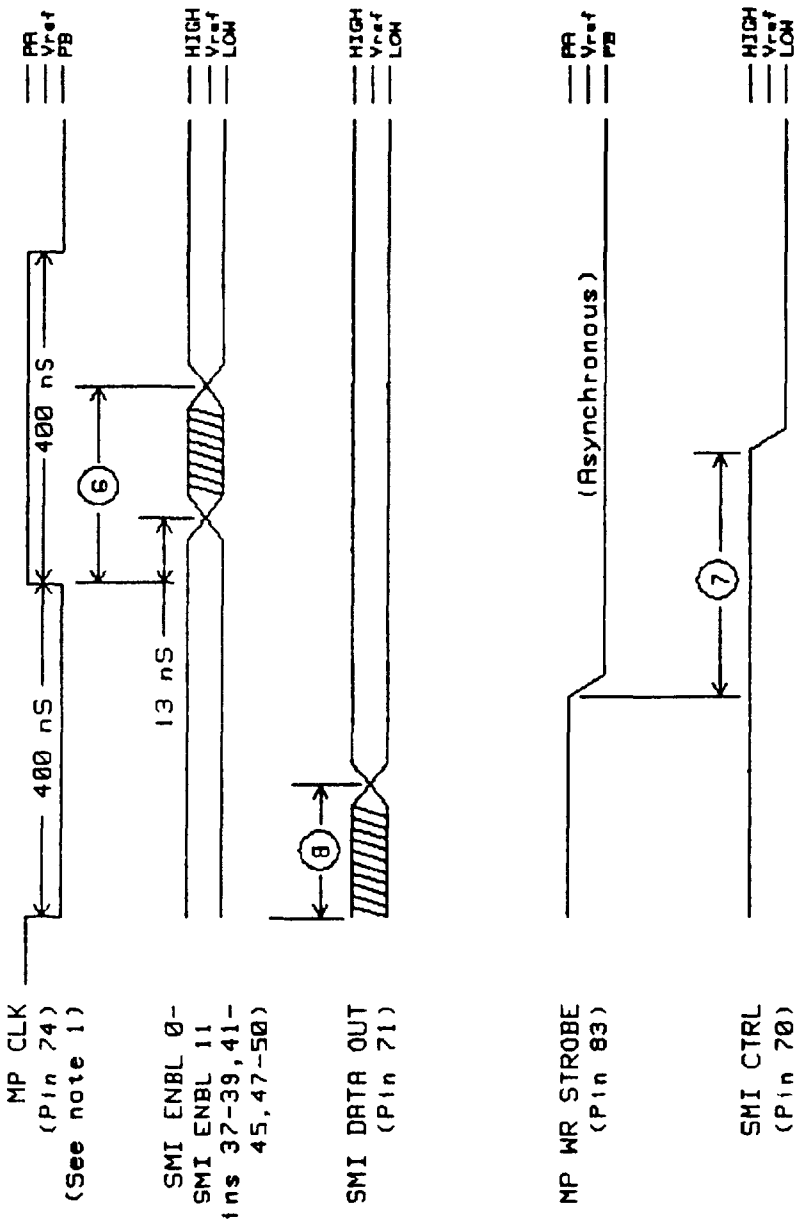


NOTE:

1. The waveform is not drawn to proportion.

FIGURE 12a. Dynamic waveforms for table XVII.

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NOTE:

1. The waveform is not drawn to proportion.

FIGURE 12b. Dynamic waveforms for table XVII.

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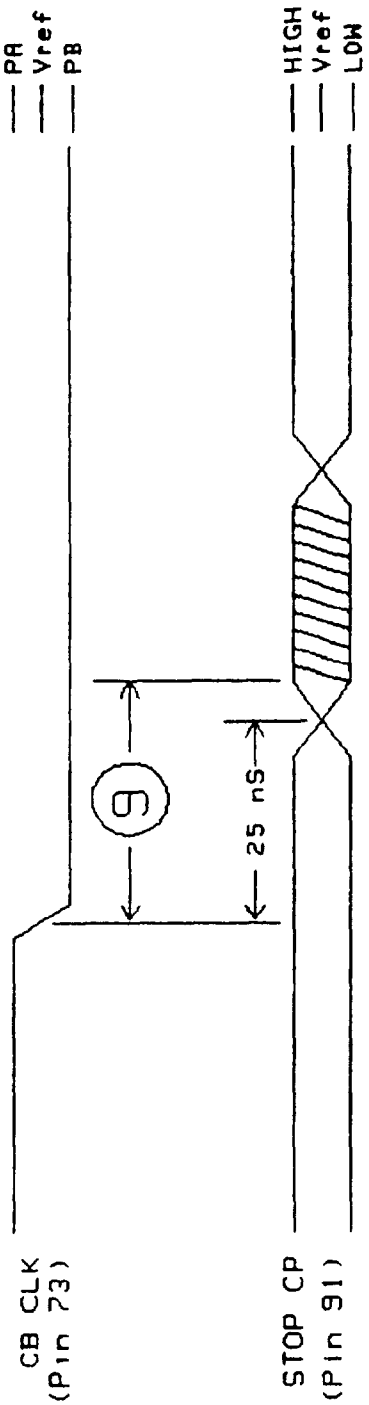


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	SCE/DESTN 15	2	2	50	pF	1-4
R2	SCE/DESTN 14	3	2	75	pF	1-4
R3	SCE/DESTN 13	4				
R4	SCE/DESTN 12	5	2	75	pF	1-4
R5	SCE/DESTN 11	6	2	50	pF	1-4
R6	SCE/DESTN 10	7	2	75	pF	1-4
R7	SCE/DESTN 9	8				
R8	SCE/DESTN 8	9				
R9	SCE/DESTN 7	11				
R10	SCE/DESTN 6	12				
R11	SCE/DESTN 5	13				
R12	SCE/DESTN 4	14				
R13	SCE/DESTN 3	15				
R14	SCE/DESTN 2	16				
R15	SCE/DESTN 1	17				
R16	SCE/DESTN 0	18	2	75	pF	1-4
R17	ENBL PC	19	8	100	pF	1-4
R18	PG SET/SEL 1	20	2	75	pF	1-4
R19	PG SET/SEL 0	21	2	75	pF	1-4
R20	MDS ADRS 0	22	2	150	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	SMI IN 11	23	2	75	pF	1-4
R22	SMI IN 10	24				
R23	SMI IN 9	26				
R24	SMI IN 8	27				
R25	SMI IN 7	28				
R26	SMI IN 6	29				
R27	SMI IN 5	30				
R28	SMI IN 4	32				
R29	SMI IN 3	33				
R30	SMI IN 2	34				
R31	SMI IN 1	35				
R32	SMI IN 0	36				
R33	MP ADDR	46	2	75	pF	1-4
R34	2 WORD READ	52	4	100	pF	1-4
R35	CP ADDR 15	53	2	75	pF	1-4
R36	CP ADDR 14	54				
R37	CP ADDR 13	55				
R38	CP ADDR 12	56				
R39	CP ADDR 11	57				
R40	CP ADDR 10	58	2	75	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	CB ADRS/WR 0	59	2	75	pF	1-4
R42	CB ADRS/WR 1	61				
R43	CB ADRS/WR 2	62	2	75	pF	1-4
R44	CB ADRS/WR 3	63	2	50	pF	1-4
R45	CB ADRS/WR 4	64	2	75	pF	1-4
R46	CB ADRS/WR 5	65				
R47	CB ADRS/WR 6	66				
R48	CB ADRS/WR 7	67				
R49	CB ADRS/WR 8	68				
R50	CB ADRS/WR 9	69				
R51	CB CLK	73				
R52	MP CLK	74	2	75	pF	1-4
R53	ENBL PH	76	4	150	pF	1-4
R54	CP CLK	77	6	100	pF	1-4
R55	BIT ENBL	79	4	100	pF	1-4
R56	CB FUNC 2	80	4	75	pF	1-4
R57	CP INSTR REF	82	4	75	pF	1-4
R58	MP WR STROBE	83	2	75	pF	1-4
R59	EN INC P CNT	84	4	75	pF	1-4
R60	CB FUNC 3	85	2	75	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	CB WR ENBL	86	2	75	pF	1-4
R62	MACLR SYNC	87	2	75	pF	1-4
R63	CB GRANT 3	88	4	100	pF	1-4
R64	CB ORIG RDY	89	2	75	pF	1-4
R65	MP RD STROBE	100	6	100	pF	1-4

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R65
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R65
Capacitance measurement	NA	± 2.4 pF	R1
	NA	± 3.65 pF	R2-R4
	NA	± 2.4 pF	R5
	NA	± 3.65 pF	R6-R16
	NA	± 4.6 pF	R17
	NA	± 3.65 pF	R18,R19
	NA	± 7.4 pF	R20
	NA	± 3.65 pF	R21-R33
	NA	± 4.8 pF	R34
	NA	± 3.65 pF	R35-R43
	NA	± 2.4 pF	R44
	NA	± 3.65 pF	R45-R52
	NA	± 7.3 pF	R53
	NA	± 4.7 pF	R54
	NA	± 4.8 pF	R55
	NA	± 3.55 pF	R56,R57
	NA	± 3.65 pF	R58
	NA	± 3.55 pF	R59
	NA	± 3.65 pF	R60-R62
	NA	± 4.8 pF	R63
	NA	± 3.65 pF	R64
	NA	± 4.7 pF	R65

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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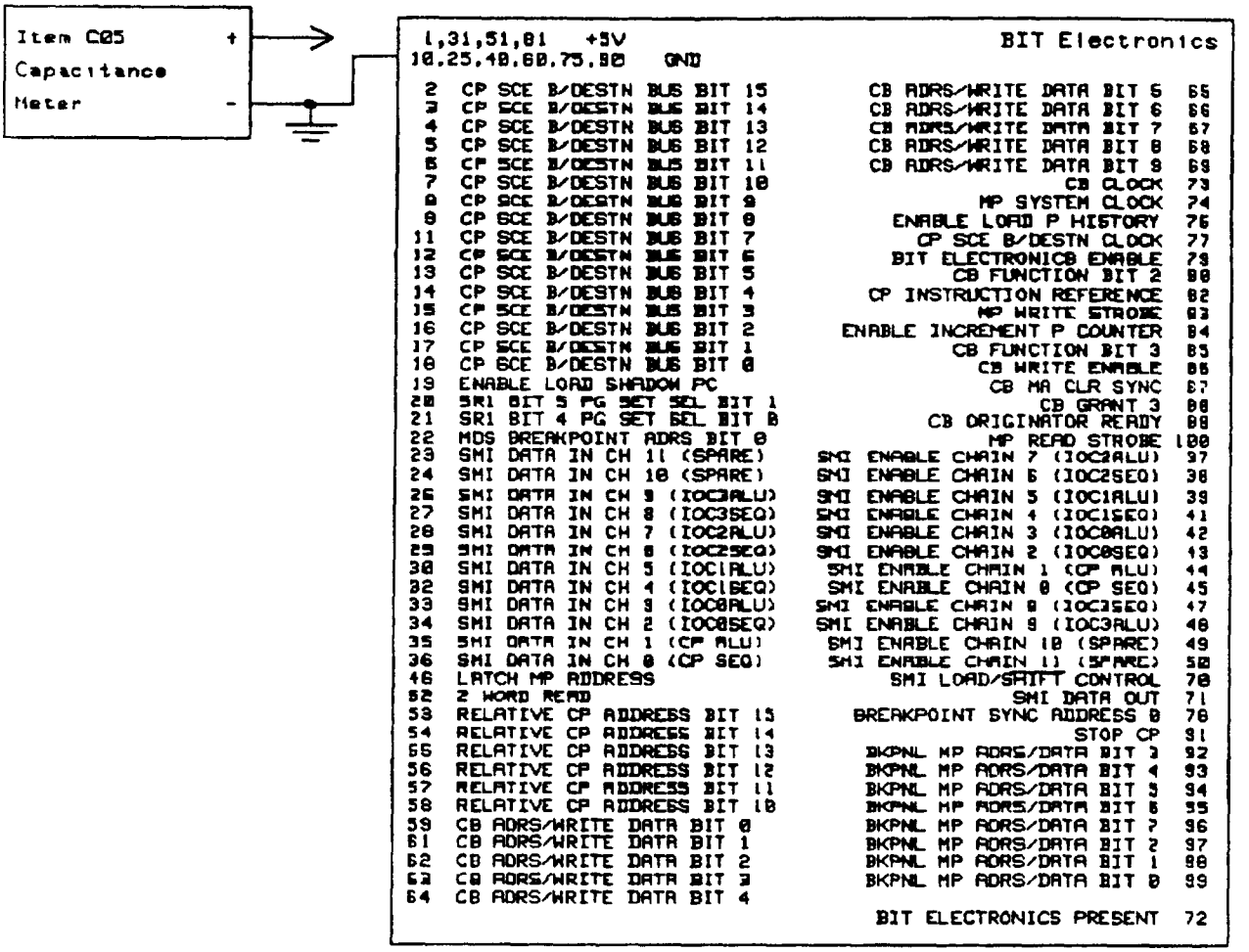


FIGURE 13. Input capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.67 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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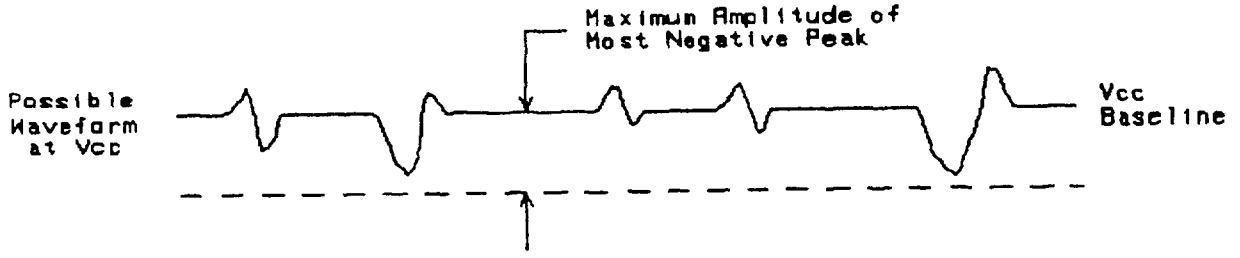
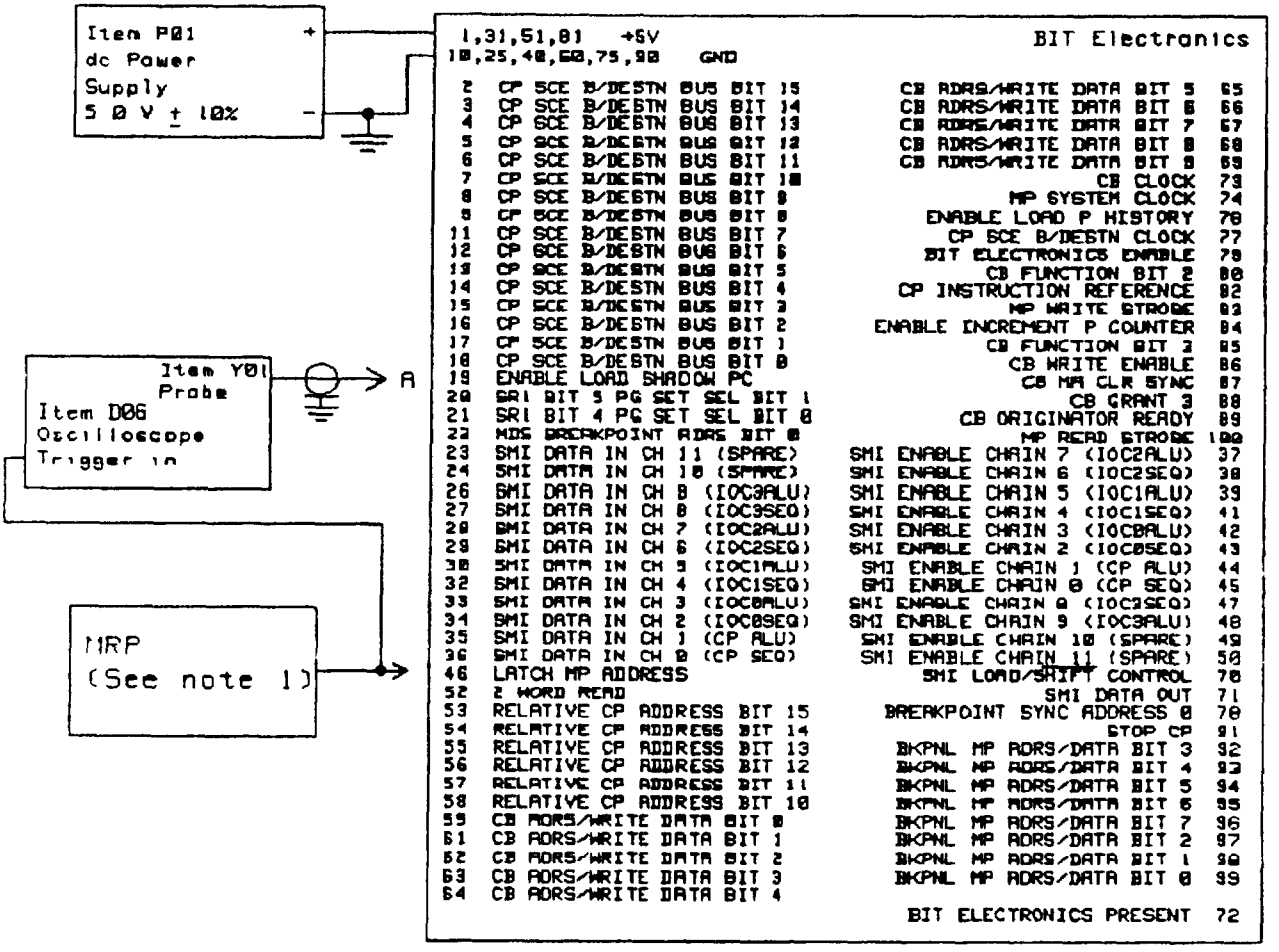


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

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